

Climate Data Management System (CDMS) Functional and Non-functional Requirements

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Expert Team on Information Management

Standing Committee on Information Management and Technology (SC-IMT)^[1]

Commission for Observation, Infrastructure and Information Systems (INFCOM)^[2]

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Chapter 1. Scope

1.1. Introduction

1.1.1. Purpose and Relationship to WMO-No. 1131

This document defines the minimum functional and non-functional requirements to support Members in establishing climate data management systems or assessing the conformance of existing systems with the *Climate Data Management System Specifications* (WMO-No. 1131, 2025 Edition).

WMO-No. 1131 provides a comprehensive framework for Climate Data Management Systems (CDMS), describing key components, governance principles, and operational practices using the classifications "Required", "Recommended", and "Optional". The high-level nature of those classifications makes it difficult in practice to assess objectively whether a given system meets the standard. This document translates the WMO-No. 1131 framework into a set of concrete requirements against which a CDMS can be assessed.

This document is primarily intended as: a tool to help WMO Members assess whether an existing or proposed CDMS meets the minimum capabilities described in WMO-No. 1131; and a reference to help developers understand the minimum capabilities expected of a conformant system. It is not a detailed technical specification or implementation guide — specific architectural and technology choices remain the responsibility of the implementing organization.

The functional requirements are expressed as requirements classes following the OGC Modular Specification approach. This specification does not place new mandatory obligations on WMO Members, and is not intended as a list of requirements that Members SHALL or SHOULD follow.

1.1.2. System Context and Boundaries

[Figure 1](#) illustrates the architectural landscape and positioning of a Climate Data Management System within the broader weather, climate, and other environmental data value chain (hereafter, climate data for conciseness). Climate data typically flows from field sensors and observing stations through national Data Collection and Processing Systems (DCPS), and subsequently to various downstream applications and services.

Key downstream systems include:

- WIS 2.0 nodes for international data exchange
- Operational meteorological systems (Numerical Weather Prediction, data assimilation, forecasting systems)
- Climate Data Management Systems for quality control and long-term climate data management
- Regional Climate Centres for regional climate monitoring and services
- Long-term preservation archives and public-facing data portals

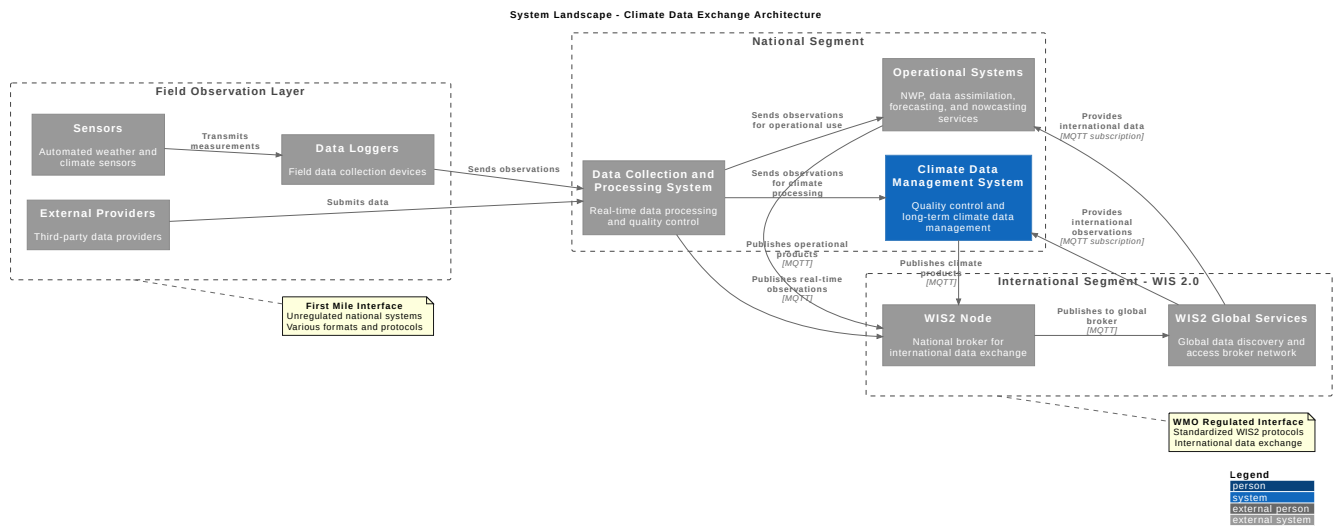


Figure 1. System landscape showing the context of Climate Data Management Systems within the data exchange architecture

Scope Boundaries:

While some operational implementations may integrate multiple components of the data flow—for example, combining DCPS and CDMS functionality within a single system—this specification focuses exclusively on the Climate Data Management System component and its specific responsibilities.

The following are explicitly OUT OF SCOPE for this specification:

- Field observation collection (sensors, data loggers, first-mile infrastructure)
- Real-time data collection and processing systems (DCPS)
- Time-critical data exchange via WIS 2.0 for operational meteorology
- Numerical Weather Prediction and forecasting systems
- General-purpose meteorological databases not specifically designed for climate data
- Operational quality control for real-time applications

The following are IN SCOPE:

- Quality control procedures specific to climate data requirements with emphasis on long-term consistency
- Detection and correction of inhomogeneities in climate time series (homogenization)
- Climate data rescue, including digitization of historical paper records
- Station metadata management with emphasis on long-term consistency and provenance
- Archival and preservation of climate datasets with appropriate documentation
- Production, publication and distribution of quality-controlled climate data products
- Integration interfaces with WIS 2.0, national networks, and international archives
- Data lineage tracking and provenance management

This specification recognizes that CDMS implementations may exist within national meteorological services (National Segment) and contribute to regional and international climate services (Regional/International Segments) as shown in Figure 1. Requirements are designed to support both contexts while maintaining flexibility for local operational arrangements.

1.2. System Architecture

1.2.1. System Context (Level 1)

Figure 2 illustrates the system context for a Climate Data Management System, showing the primary users, the CDMS boundary, and its interactions with external systems. The diagram identifies key integration points including:

- **Data sources:** The national Data Collection and Processing System (DCPS), and WIS 2.0 Global Services (via national WIS2 Node).
- **Users:** Climate scientists, data managers, and system administrators
- **Downstream systems:** Long-term archives, climate data stores, and climate analysis platforms
- **Support systems:** Identity management and operations monitoring

While these external systems and their interactions are shown for context, their internal specifications are outside the scope of this document. This specification focuses exclusively on the requirements for the CDMS component itself and its standardized interfaces with these external systems.

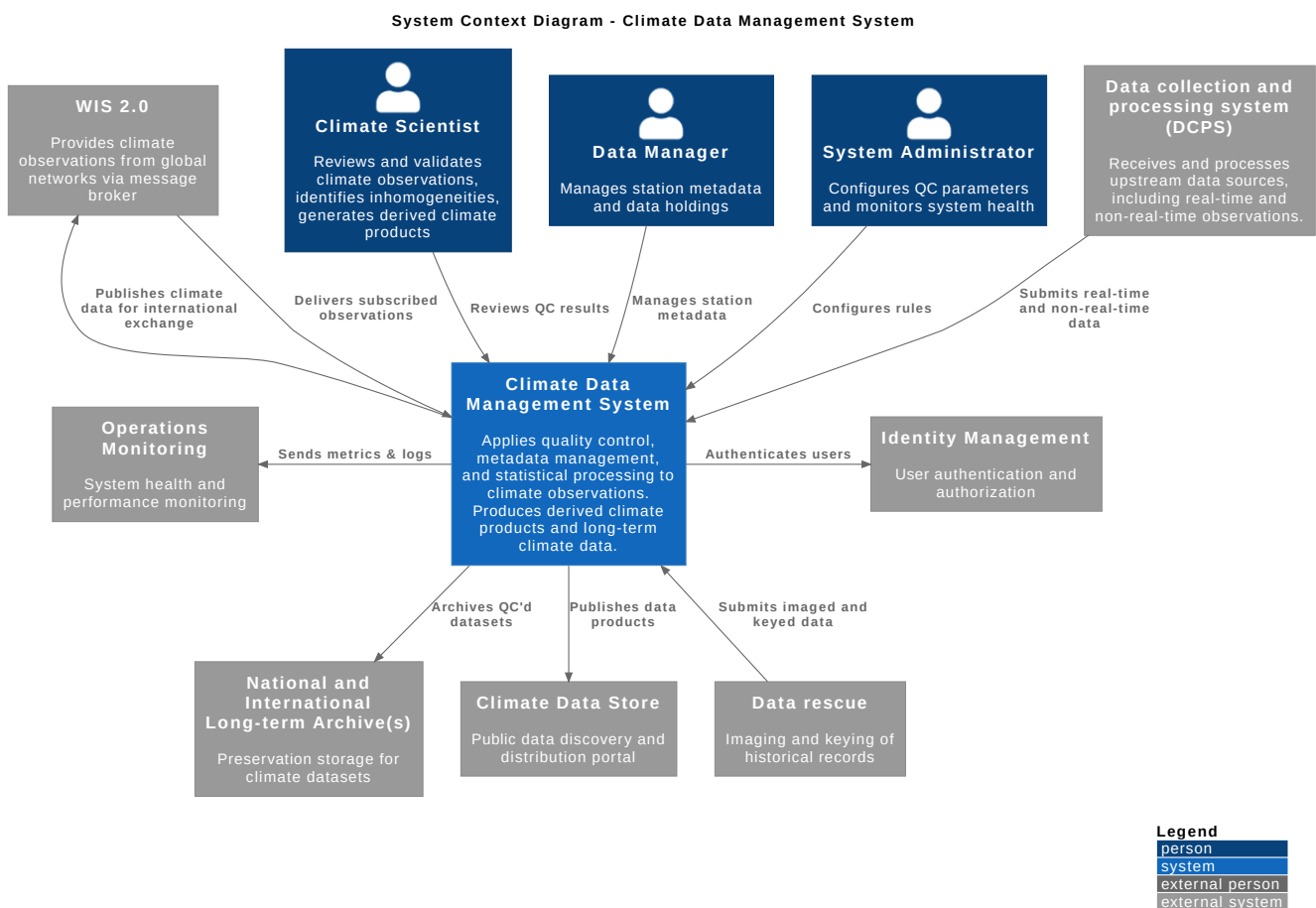


Figure 2. CDMS system context diagram showing users and external system interfaces

1.2.2. Container Architecture (Level 2)

Figure 3 provides a high-level view of the major functional components and data stores within the Climate Data Management System. This diagram is intentionally technology-agnostic, allowing implementers flexibility in architectural and technology choices while conforming to the functional requirements of this specification.

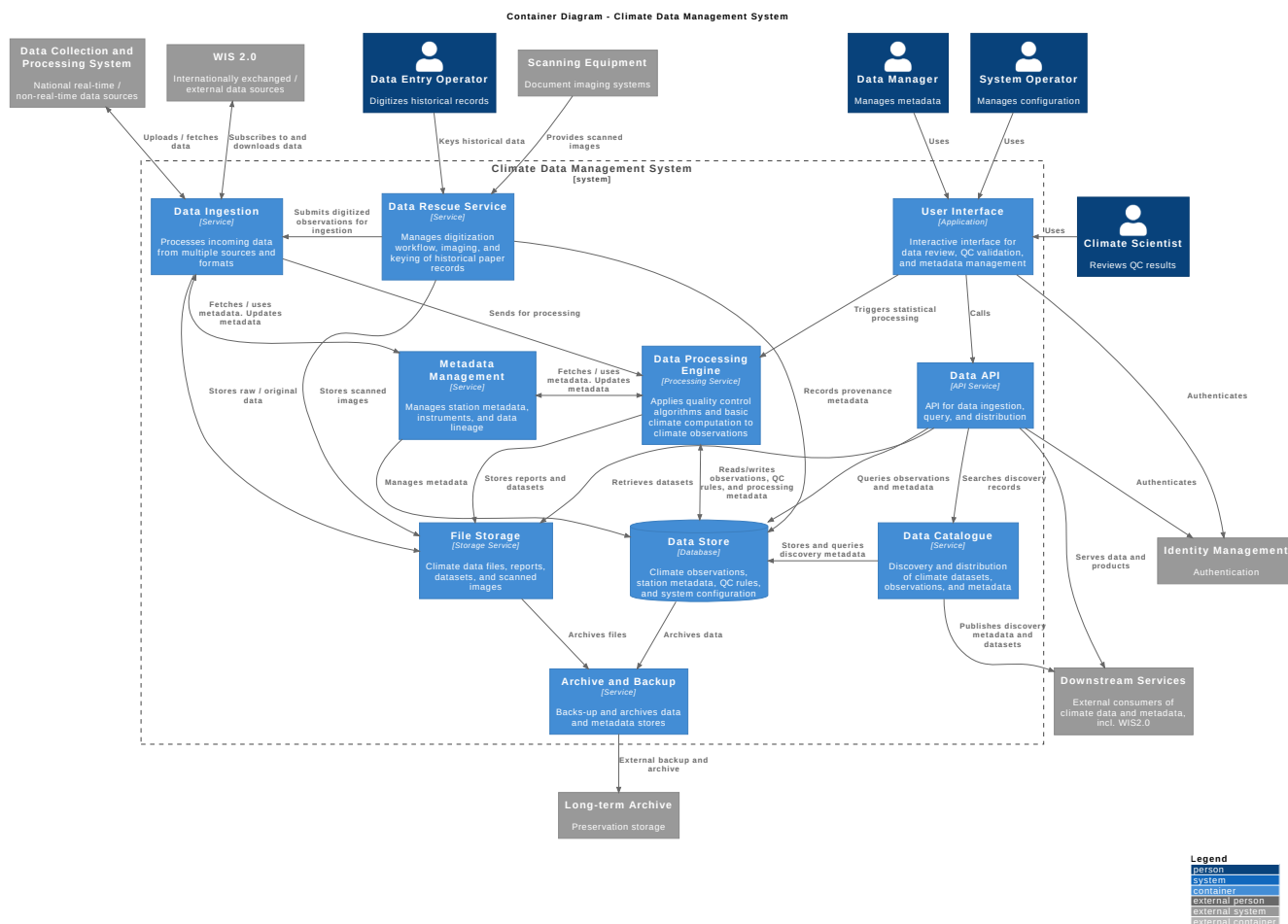


Figure 3. Container diagram showing major functional components and data stores

Key Containers:

- **User Interface:** Provides interactive capabilities for climate scientists, data managers, and system administrators to review quality-controlled data, validate observations, manage station metadata, configure system parameters, and monitor system operations. This may be implemented as a web application, desktop application, or a combination of both.
- **Data API:** Exposes standardized interfaces for data ingestion from external sources, data query and retrieval by users and downstream systems, and publication of climate data products. The API should support international standards for climate data exchange including OGC API standards.
- **Data Ingestion:** Processes incoming observations from the national DCPS (national real-time and non-real-time data) and from WIS 2.0 (international observations), as well as digitized historical observations submitted by the Data Rescue Service. Handles multiple data formats and protocols, performing initial validation and normalization.
- **Data Processing Engine:** Applies quality control algorithms to climate observations including range checks, temporal consistency checks, spatial consistency checks, and climatological validation. Also performs statistical analysis to detect and adjust for non-climatic discontinuities in observation time series (homogenization), producing both adjusted data and comprehensive documentation of the process.
- **Metadata Management:** Maintains comprehensive station and element metadata including location information, instrumentation history, observing practices, element definitions (units, codes, and controlled vocabularies), and data lineage. Supports both data ingestion validation and quality control processing, and ensures compliance with international metadata standards.
- **Data Rescue Service:** Manages the digitization workflow for historical climate records, including coordination of document imaging, data entry from paper records, and quality assurance of digitized

observations. Preserves both the scanned images and the keyed data with full provenance information. Digitized observations flow through the same Data Processing Engine as contemporary observations to ensure consistent quality standards.

- **Data Catalogue:** Maintains persistent discovery metadata records describing available datasets, observations, and metadata holdings. Supports search and discovery via the Data API and publishes discovery metadata to downstream services.
- **Archive and Backup:** Manages scheduled backup and archival of the Data Store and File Storage contents. Submits datasets to external long-term preservation archives and maintains backup copies to support disaster recovery.

Data Stores:

- **Data Store:** Persistent storage for climate observations with their associated quality flags and processing history, station and element metadata (including units, codes, and controlled vocabularies), quality control rules and thresholds, system configuration, homogenization parameters, and data lineage information. Maintains historical records of all metadata changes to support long-term climate analysis.
- **File Storage:** Manages climate-standard file formats (such as NetCDF-CF), quality control reports, homogenization analysis outputs, digitized document images, and archived datasets. Supports versioning and long-term preservation requirements.

External System Integration:

The system integrates with external systems at well-defined conformance boundaries where standardized protocols and data formats apply:

- **WIS 2.0:** International data exchange using MQTT messaging and standardized data formats for both consuming international observations and publishing national climate products.
- **National Data Collection and Processing System:** Real-time and non-real-time data sources and processing, including connection to data loggers, sensors and other external data sources.
- **Long-term Preservation Archives:** Climate dataset archival using standard formats.
- **Downstream Services:** External consumers of climate data and metadata, including climate data stores, analysis platforms, and WIS 2.0 nodes, accessed via the Data API and Data Catalogue.
- **Identity Management Systems:** Enterprise authentication and authorization using standard protocols (LDAP, OAuth 2.0, SAML).

This architectural decomposition supports the modular requirements structure of this specification, where different containers may be subject to different requirements classes based on their functional responsibilities.

1.3. Document Structure and Organization

To facilitate clear understanding and practical application, this document is structured into the following key sections:

- **Functional Requirements:** Details what a CDMS shall do, providing specific, granular requirements for core functionalities such as data ingestion and validation, transformation and storage, metadata management, quality control, data rescue, processing, and user interface interactions. These requirements operationalize the high-level functional components outlined in WMO-No. 1131.
- **Non-Functional Requirements:** Defines how well a CDMS shall perform, covering essential attributes like performance, scalability, reliability, maintainability, usability, interoperability, data governance, and comprehensive security measures. These requirements ensure that a CDMS not only fulfills its

functional duties but also operates effectively, securely, and sustainably in various environments.

- **Appendix:** Provides supplementary information, including a glossary of terms, a list of acronyms used throughout the document, references to related standards and specifications, and any other relevant supporting material that aids in the understanding and application of these specifications.

Each requirement within the functional and non-functional sections is designed to be clear, assessable, and directly applicable, serving as a valuable reference for both system development and conformance assessment activities in the context of WMO-No. 1131.

[1] <https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt>

[2] <https://community.wmo.int/governance/commission-membership/infcom>

Chapter 2. Conformance

This specification defines the following conformance classes corresponding to the functional capabilities of a Climate Data Management System:

- **Core Data Management** (Required): Foundational observation storage and retrieval, provenance metadata management, and basic station and sensor metadata storage. This class provides the data management foundation on which all other conformance classes depend.
- **Quality Control** (Required): Climate-specific QC procedures including range, temporal, spatial, and climatological checks
- **Data Ingestion** (Required): Multi-format data import, validation, and normalization
- **Data API** (Required): Standards-based data query, retrieval, and distribution interfaces
- **Basic Climate Computation** (Required): Computation of derived meteorological variables, summary statistics, and homogenized series from quality-controlled observations
- **Data Rescue** (Optional): Digitization workflow for historical paper records including imaging and data entry
- **Metadata Management** (Required): Advanced station and element metadata management including WIGOS compliance, complete station history, discovery metadata, and metadata quality validation
- **User Interface** (Required): Interactive visualization, data review, and quality control validation capabilities
- **Climate Products** (Recommended): Production and publication of standard climate products and indices
- **Processing Automation** (Recommended): Scheduled and automated execution of routine data processing workflows

NOTE

The classification of conformance classes as Requirement, Recommendation and Permission aligns with the Required, Recommended and Optional classifications respectively from WMO-No. 1131. Required classes must be implemented for a system to be considered conformant. Recommended classes should be implemented where operationally feasible. Optional classes may be implemented based on specific organizational needs.

Chapter 3. References

3.1. Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document.

3.1.1. WMO Publications

- WMO-No. 8: *Guide to Meteorological Instruments and Methods of Observation*. World Meteorological Organization, Geneva.
- WMO-No. 49: *WMO Technical Regulations*, Volume I: General Meteorological Standards and Recommended Practices. World Meteorological Organization, Geneva.
- WMO-No. 306: *Manual on Codes*, Volume I.1 (Alphanumeric data representation) and Volume I.2 (Binary codes, including BUFR). World Meteorological Organization, Geneva.
- WMO-No. 488: *Guide to the Global Observing System*. World Meteorological Organization, Geneva.
- WMO-No. 1060: *Manual on the WMO Information System*, Volume II: Approved WIS Technical Specifications (including the WMO Core Metadata Profile version 2, WCMP2). World Meteorological Organization, Geneva.
- WMO-No. 1131: *Climate Data Management System Specifications* (2025 Edition). World Meteorological Organization, Geneva.
- WMO-No. 1160: *Manual on the WMO Integrated Global Observing System*. World Meteorological Organization, Geneva.
- WMO-No. 1186: *World Weather Records*. World Meteorological Organization, Geneva.
- WMO-No. 1192: *WIGOS Metadata Standard*. World Meteorological Organization, Geneva.
- WMO-No. 1203: *WMO Guidelines on the Calculation of Climate Normals*. World Meteorological Organization, Geneva.
- WMO-No. 1238: *Manual on High-quality Global Data Management Framework for Climate*. World Meteorological Organization, Geneva.
- WMO-No. 1269: *Guidelines on Surface Station Data Quality Control and Quality Assurance for Climate Applications*. World Meteorological Organization, Geneva.

3.1.2. IETF

- RFC 2119: Bradner, S. (1997): *Key words for use in RFCs to Indicate Requirement Levels*. Internet Engineering Task Force. <https://www.rfc-editor.org/rfc/rfc2119>
- RFC 4180: Shafranovich, Y. (2005): *Common Format and MIME Type for Comma-Separated Values (CSV) Files*. Internet Engineering Task Force. <https://www.rfc-editor.org/rfc/rfc4180>

3.1.3. W3C

- PROV-DM: Moreau, L., and P. Missier (eds.) (2013): *PROV-DM: The PROV Data Model*. W3C Recommendation, 30 April 2013. <https://www.w3.org/TR/prov-dm/>
- PROV-O: Lebo, T., et al. (eds.) (2013): *PROV-O: The PROV Ontology*. W3C Recommendation, 30 April 2013. <https://www.w3.org/TR/prov-o/>

3.1.4. CF Conventions

- Hassell, D., et al.: *NetCDF Climate and Forecast (CF) Metadata Conventions*. <https://cfconventions.org/>

3.1.5. OGC

- OGC 15-078r6: *OGC SensorThings API Part 1: Sensing*. Open Geospatial Consortium.
- OGC 17-069r4: *OGC API — Features — Part 1: Core*. Open Geospatial Consortium.
- OGC 19-086r6: *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium.
- OGC 20-004r1: *OGC API — Records — Part 1: Core*. Open Geospatial Consortium.

3.2. Informative references

The following documents are cited for context and guidance. They do not form part of the normative requirements of this specification.

- WMO-No. 49: *WMO Technical Regulations*, Volume I — cited in WMO-No. 1238 in relation to observations metadata standards.
- WMO-No. 1182: *Guidelines on Good Practices for Data Rescue*. World Meteorological Organization, Geneva.
- ISO 19115 family: *Geographic information — Metadata*. International Organization for Standardization. Readers familiar with ISO 19115 will recognize concepts used in this specification; ISO standards are not freely available and are not normative requirements of this document.

Chapter 4. Terms and definitions

For the purposes of this document, the terms and definitions given in WMO-No. 1131 and the following apply:

Availability

proportion of time a system or service is operational and accessible to users, typically expressed as a percentage over a defined measurement period

NOTE

In this specification, availability refers to data retrieval and ingestion services. Not to be confused with data availability, which refers to the presence or absence of observation records in the archive.

Break point

a point in time at which a statistically significant discontinuity is detected in a climate observation time series, attributable to a non-climatic cause such as station relocation, instrument change, or change in observation practice

NOTE

Break points are identified during the homogenization process. Each break point separates segments of a time series within which different adjustment factors may be applied.

Climate normals

standard reference values representing the average state of a climate variable over a defined reference period, computed in accordance with WMO guidelines (WMO-No. 1203). Standard reference periods are 30 consecutive years (currently 1991–2020)

Data Collection and Processing System (DCPS)

a national system responsible for the real-time collection, initial validation, and transmission of observational data from field sensors and observing networks to downstream data management systems

NOTE

The DCPS typically applies real-time quality checks appropriate for operational meteorology. The primary integration point addressed by this specification is the delivery of data from the DCPS to the Climate Data Management System for long-term quality assurance and archival. The internal operation of the DCPS is outside the scope of this specification.

Data lineage

a summary record of the processing history of a dataset as a whole, documenting the sequence of operations applied from source data through to the current published state, structured following the W3C PROV Data Model. Distinct from observation-level provenance metadata, which records the immutable processing history of individual observations

Data rescue

the process of recovering historical climate observations from non-digital sources — primarily paper records, registers, and ledgers — by transcribing the data into machine-readable form, with appropriate quality assurance, and integrating the rescued observations into the observational record with complete provenance

NOTE

Automated digitization via optical character recognition (OCR) is outside the scope of this specification.

Data stewardship

the assignment to an entity of defined rights and responsibilities for acquiring, managing, maintaining, and ensuring the long-term accessibility and integrity of climate data, encompassing quality assurance, lifecycle management, access control, and accountability for the ongoing fitness of the data record. Adapted from WMO-No. 1238

Derived observation

an observation value computed by the system from one or more source observations by means of a documented formula, aggregation, or statistical procedure, as distinct from a value directly reported by a measuring instrument. Examples include computed meteorological variables (e.g. mean sea level pressure derived from station pressure), statistical aggregates (e.g. daily maximum temperature), and homogenized values

Discovery metadata

dataset-level metadata records that make data findable in a discovery catalogue, describing what data exists, where it can be accessed, its spatial and temporal extent, responsible organization, and data policy classification

NOTE

Distinct from station and sensor metadata, which describe the observing platform and instruments that produced the data.

Double-key entry

a data entry quality assurance procedure in which two operators independently transcribe the same source document, after which the two transcriptions are compared automatically and any discrepancies are flagged for resolution before the values are accepted into the observational record

Earth system domains

the atmospheric, terrestrial, hydrological, cryospheric, and oceanic environments from which in-situ point observations relevant to climate are collected

Fitness-for-purpose

an assessment of whether a dataset meets the quality and completeness requirements of a particular intended application or use case, summarizing quality metrics — including completeness, temporal continuity, quality control flag distribution, and known systematic issues — in terms of suitability for defined use cases such as climatological trend analysis, climate normals computation, or extreme event studies

Homogenization

the process of detecting and correcting non-climatic discontinuities in long-term observation time series caused by changes in station location, instrumentation, observation practices, or surrounding environment, in order to produce a temporally consistent series suitable for climate trend analysis. Results in an adjusted series stored alongside the original, unadjusted series with full provenance documentation

Inhomogeneity

a discontinuity or step-change in a climate time series attributable to non-climatic factors such as station relocation, instrument replacement, changes in observation practices, or changes to the station environment, rather than to actual changes in the observed climate variable

In-situ point observation

an observation obtained by direct measurement at a specific geographic location, representing conditions at that point in space

NOTE

Encompasses measurements from fixed stations, where geographic location is derived from station metadata, and from mobile platforms such as ships, buoys, radiosondes, and aircraft, where geographic location is stored with each individual observation. Excludes gridded, raster, reanalysis, satellite-derived, radar composite, and forecast data products. The scope of this specification is restricted to in-situ point observations.

Normalization

the process of transforming observational data from diverse source formats, encodings, and unit systems into a common internal representation, resolving differences in format conventions, units of measure, coordinate systems, timestamp formats, and character encodings

NOTE

Not to be confused with statistical normalization, which scales data to a standard range or distribution.

Observable element

a uniquely defined measurable quantity characterized by its measurement class (e.g. temperature, precipitation, pressure), accumulation or averaging duration, statistical function (instantaneous, mean, maximum, minimum, accumulated total), and unit of measure

NOTE

Two measurements that differ in any of these properties constitute distinct elements and are stored and described separately to ensure unambiguous interpretation across stations and datasets. Source data provenance (e.g. transmission format or originating network) is recorded separately rather than embedded in the element definition.

Provenance metadata

information documenting the origin, processing history, and all modifications applied to an observation or dataset throughout its lifecycle, from initial acquisition through archival and distribution. For individual observations, provenance includes source identification, ingestion details, quality control decisions, manual corrections, homogenization adjustments, and data rescue workflows. Adapted from WMO-No. 1238

Quality flag

a coded indicator attached to an observation record documenting the outcome of one or more quality control checks applied to that observation, recording whether a value is accepted, suspect, or rejected, and indicating the specific check that assigned the flag and the level at which it was applied (automatic, manual, or final)

Chapter 5. Conventions

For the purposes of this document, the conventions described in this clause apply.

5.1. Identifiers

The normative provisions in this module are denoted by URIs which follow a hierarchical structure to ensure unique identification and traceability.

The base URI for this specification is: `spec-name/[version]/[module-name]`

All requirements that appear in this document are denoted by **partial URIs** which are relative to this base.

The structure for partial URIs is defined as follows:

5.1.1. Requirements URIs

Requirements are identified by a URI fragment that includes the Requirements Class name and a unique identifier for the specific requirement.

- **Format:** `/req/[requirements-class-name]/[requirement-id]`
- **Example:** `/req/quality-control/range-check`

5.1.2. Recommendation URIs

Recommendations are identified by a URI fragment that includes the Recommendations Class name and a unique identifier for the specific requirement.

- **Format:** `/rec/[recommendation-class-name]/[recommendation-id]`
- **Example:** `/rec/climate-computation/homogenization-detection`

5.2. Use of IETF RFC 2119 Keywords

The keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in IETF RFC 2119.

The keywords indicate the level of strictness required for compliance with a normative provision:

- **MUST/REQUIRED/SHALL:** These keywords mean that the definition is an **absolute requirement** of the specification.
- **MUST NOT/SHALL NOT:** These phrases mean that the definition is an **absolute prohibition** of the specification.
- **SHOULD/RECOMMENDED:** These keywords mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
- **SHOULD NOT/NOT RECOMMENDED:** These phrases mean that there may exist valid reasons in particular circumstances when the behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before proceeding.
- **MAY/OPTIONAL:** These keywords mean that an item is truly optional.

5.3. Typographical and Notational Conventions

The following typographical conventions are used throughout this document to distinguish different types of content:

- **Monospace Text**: Used for source code examples, file names, directory names, machine-readable data structures (e.g., JSON keys, XML tags), URIs, and literal text that appears in a program or command line.
- **Italics**: Used for titles of documents, variable placeholders within code or URI templates, and for emphasis.
- **[Bracketed] Text**: Used for placeholders in this template (e.g., `[spec-name]`, `[module-name]`) that must be replaced with specific content during implementation.
- **[NOTE], [TIP], [IMPORTANT]**: Used to highlight informative text blocks that provide context, advice, or warnings that are **not** part of the normative requirements.

Chapter 6. Functional Requirements

6.1. Requirements Class: Core

Requirements class	/req/core
Target type	Software system
Dependency	None

6.1.1. Overview

The Core requirements class defines the requirements for basic observation storage, retrieval, and metadata management capabilities. This provides the foundational data management capabilities for a Climate Data Management System and support for long-term data stewardship.

6.1.2. Requirements

6.1.2.1. Observation Data Model

Requirement 1	
Identifier	/req/core/data-model
Statement	The system SHALL implement a data model capable of storing in-situ point observations from across the Earth system domains and disciplines. The data model SHALL include: • Station and sensor metadata • Observed element/variable • Observation value/result and associated units • Quality flag status • Observation timestamp, including time zone support • Geographic location of the observation (derived from station metadata for fixed stations; stored per-observation for mobile platforms) • Provenance metadata

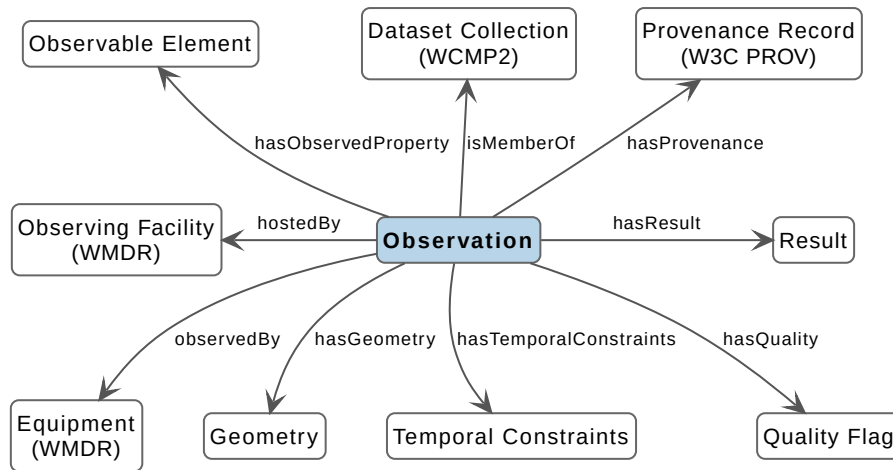


Figure 4. Observation conceptual model

NOTE

The model is consistent with the WMO Core Climate Data Model – Observations (WCCDM-OBS, FM 251), proposed to INFCOM-3 and still under experimental status (JSON schema: <https://schemas.wmo.int/wccdm-obs/2024/wccdm-obs.json>). WCCDM-OBS is not a normative reference of this specification; further alignment is anticipated as the model progresses towards formal adoption.

6.1.2.2. Observation Storage

Requirement 2	
Identifier	/req/core/observation-storage
Statement	The system SHALL provide persistent storage for observation records.

6.1.2.3. Observation Retrieval

Requirement 3	
Identifier	/req/core/observation-retrieval
Statement	The system SHALL support retrieval of observations filtered by: • Station identifier or group of stations • Geographic bounding box or polygon • Time period (start and end timestamps) • Observed element/variable • Quality control status

6.1.2.4. Station and Sensor Metadata Storage

Requirement 4	
Identifier	/req/core/metadata-storage
Statement	
A	The system SHALL provide persistent storage for station and sensor metadata as specified in /req/metadata-management/wigos.

B	The system SHALL support extensibility mechanisms to store metadata fields not covered by the WIGOS Metadata Standard, including national identifiers, legacy identifiers from predecessor networks, and other organization-specific metadata attributes.
---	---

6.1.2.5. Station and Sensor Metadata Retrieval

Requirement 5	
Identifier	/req/core/metadata-retrieval
Statement	The system SHALL support retrieval of station and sensor metadata filtered by: • Station identifier (WIGOS Station Identifier, national identifier, or legacy identifier) • Geographic extent (bounding box, polygon, or proximity search) • Administrative boundaries (country, region, organization) • Station classification or network affiliation • Observed element capability • Operational status (active, historical, planned) • Temporal validity (metadata state at a specified point in time)

6.1.2.6. Station and Sensor Metadata Versioning

Requirement 6	
Identifier	/req/core/metadata-versioning
Statement	The system SHALL maintain temporal versioning of all station and sensor metadata such that: • All changes to metadata are recorded with effective date and time • Historical states of metadata can be reconstructed for any point in time • The system can return metadata valid at the time of any given observation • Metadata changes do not overwrite or destroy previous versions

6.1.2.7. Provenance Metadata Storage

Requirement 7	
Identifier	/req/core/provenance-storage

Statement	The system SHALL store provenance metadata for all observation data documenting the complete data lineage including: • Original data source identification (network, station, transmission method) • Acquisition timestamp (when data entered the system) • Ingestion processing details (format conversion, unit normalization, coordinate transformation) • Quality control processing history (quality control procedures applied, flags assigned, timestamps) • Manual interventions (edits, corrections, deletions with operator identification and justification) • Derived data lineage (source observations, algorithms, parameters used in derivation) • Homogenization provenance: break points detected with dates and statistical confidence levels; adjustment factors applied to each segment; reference series used; statistical significance of detected breaks; correspondence with known station changes; homogenized values stored as derived observations with both original and adjusted series retained • Data rescue provenance (source document reference, image identifier, keying method, verification status)
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6.1.2.8. Provenance Metadata Retrieval

Requirement 8	
Identifier	/req/core/provenance-retrieval
Statement	The system SHALL support retrieval of provenance metadata including: • Complete processing history for any observation or set of observations • Identification of all transformations applied to data from acquisition to current state • Audit trail queries by operator, time period, or processing type • Traceability from derived products back to original source observations • Export of provenance information in machine-readable format (e.g., W3C PROV-O, ISO 19115 lineage)

6.1.2.9. Provenance Metadata Immutability

Requirement 9	
Identifier	/req/core/provenance-immutability
Statement	The system SHALL ensure provenance records are immutable such that: • Provenance entries cannot be modified or deleted after creation • Original (raw) observation values are preserved even when corrections are applied • A complete audit trail exists for all data modifications

6.1.2.10. Batch-processing

Requirement 10	
Identifier	/req/core/batch-processing
Statement	The system SHALL support batch operations for efficiency: • Bulk station metadata updates via file upload • Batch quality control flag modifications with justification • Bulk data export for multiple stations and time ranges • Progress indicators for long-running operations • Ability to cancel or pause long-running batch operations

6.2. Requirements Class: Observations Quality Control

Requirements class	/req/quality-control
Target type	Software system
Dependency	/req/core

6.2.1. Overview

The Observations Quality Control requirements class defines requirements to ensure observation data meets quality standards for long-term environmental records through systematic quality checks that detect erroneous, suspicious, or inconsistent values while maintaining complete audit trails of all quality control decisions. Quality control is essential for establishing trustworthy observational records suitable for long-term trend analysis and monitoring. The results of quality control processing may also help identify systemic error patterns in the observations that imply the need for remedial training or other actions (see /req/data-governance/stewardship), and/or may inform design changes to the quality control algorithms if too few errors are detected or too many values are flagged as suspect.

6.2.2. Requirements

6.2.2.1. Observation Retention

Requirement 11	
Identifier	/req/quality-control/observation-retention
Statement	The system SHALL retain all observation records regardless of quality flag status. Observations that fail quality control checks SHALL remain in the system with their assigned quality flags and the complete record of checks applied.

6.2.2.2. Quality Flag Management

Requirement 12	
Identifier	/req/quality-control/flags

Statement	The system SHALL maintain quality flags and: • Support multiple quality control levels (automatic, manual, final) • Preserve quality control history and audit trail • Allow manual override with justification • Record which specific quality control check(s) triggered each flag and the reason for failure
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Recommendation 1	
Identifier	/rec/quality-control/flagging-convention
Statement	The system SHOULD adopt a published quality control flagging convention to facilitate data exchange and interoperability. Applicable references include the <i>Guide to the Global Observing System</i> (WMO-No. 488), WMO-No. 1269, and the <i>Guide to Climatological Practices</i> (WMO-No. 100).

6.2.2.3. Quality Control Configuration Management

Requirement 13	
Identifier	/req/quality-control/configuration
Statement	The system SHALL support configuration of quality control parameters, including: • Station-specific and element-specific thresholds • Seasonal variation of thresholds where climatologically appropriate • Enabling/disabling of individual checks per station or element • Version control of quality control rule configurations with effective dates

NOTE

Thresholds may be either fixed values (e.g. absolute physical limits) or dynamically derived (e.g. based on climatological statistics such as percentiles or standard deviations). Simple fixed-threshold configurations should remain straightforward to set up; advanced options such as seasonally varying or statistically derived thresholds are available but not required for basic operation.

6.2.2.4. Quality Control Execution

Requirement 14	
Identifier	/req/quality-control/execution
Statement	The system SHALL support quality control processing initiated in multiple ways: • Automatically upon data ingestion • On demand by an authorized user, including reprocessing of previously checked observations • As a scheduled or batch process, including reprocessing of previously checked observations

NOTE

A series of standard checks is defined in the following sections. Not all checks are applicable to all variables or observation frequencies; for instance, rate of change and

spike checks are not relevant to the quality control of daily accumulated rainfall.

6.2.2.5. Range Checks

Requirement 15	
Identifier	/req/quality-control/range-checks
Statement	The system SHALL perform range validation checks, including: • Gross error limits (physically impossible values, e.g. wind direction greater than 360 degrees or temperature greater than 80°C) • Climatologically-based range checks • Sensor-based range checks (measurements within observable range for sensor) • Configurable thresholds per element and station

NOTE

Element-specific gross error limits are defined in the *Guide to the Global Observing System* (WMO-No. 488) and the *Guide to Meteorological Instruments and Methods of Observation* (WMO-No. 8) and should be used as the basis for gross error limit checks where available.

6.2.2.6. Temporal Consistency Checks

Requirement 16	
Identifier	/req/quality-control/temporal-checks
Statement	The system SHALL perform temporal consistency checks, including: • Flatline checks - detection of repeated identical values • Rate of change checks - validation of temporal gradients • Spike checks - detection of anomalous rapid changes followed by rapid reversals

6.2.2.7. Spatial Consistency Checks

Requirement 17	
Identifier	/req/quality-control/spatial-checks
Statement	The system SHALL perform spatial consistency checks including: • Comparison with neighbouring observations using configurable selection criteria, including maximum distance, elevation difference, station type, covariability between stations, and predefined station lists • Spatial regression tests

NOTE

Where no neighbouring stations exist within a practicable distance — as is the case for many island and remote stations — spatial consistency checks cannot be applied. The maximum distance threshold used for neighbour selection should be configurable to reflect regional station density.

6.2.2.8. Internal Consistency Checks

Requirement 18	
Identifier	/req/quality-control/internal-consistency
Statement	The system SHALL perform internal consistency checks to validate physical relationships between simultaneously observed elements as specified in WMO-No. 1269.

NOTE

Examples of internal consistency checks include validation of temperature relationships ($T_{\min} \rightarrow T_{\text{avg}} \rightarrow T_{\max}$), humidity constraints, pressure consistency, and precipitation-weather code agreement. WMO-No. 1269 provides comprehensive guidance on element-specific checks.

6.3. Requirements Class: Data Ingestion

Requirements class	/req/data-ingestion
Target type	Software system
Dependency	/req/core

6.3.1. Overview

The Data Ingestion requirements class defines requirements for the reliable ingestion of observational data from diverse sources and formats. This requirements class is restricted to observational (in-situ) data from stations and does not cover the ingestion of gridded or raster data products (e.g., satellite imagery, reanalysis fields, radar composites). Data ingestion serves as the entry point for observations into the CDMS or climate record, requiring careful handling to preserve data integrity through validation and normalization while maintaining complete provenance of source data characteristics.

NOTE

This requirements class covers ingestion via automated electronic data streams and manual file upload. Data entry via keyboard transcription of historical paper records is addressed in /req/data-rescue. A CDMS may serve both operational climate product generation and the long-term climate archive; the ingestion requirements in this class apply in both contexts.

6.3.2. Requirements

6.3.2.1. Automated Data Collection

Requirement 19	
Identifier	/req/data-ingestion/automated-collection
Statement	

A	The system SHALL support the following data collection triggers: • Scheduled polling: Configurable polling of external data sources at defined intervals (e.g., automated weather station (AWS) networks) • Real-time subscription: Subscription to message brokers (e.g., MQTT) for event-driven data collection • File system monitoring: Automatic detection and ingestion of new files in monitored directories
B	The system SHALL support the following data transfer protocols: • FTP and SFTP (plain FTP should only be used within secure internal networks; SFTP or FTPS is required for internet-facing transfers) • HTTP/HTTPS • MQTT (for WIS 2.0 message broker integration) • RESTful API calls to external observation systems with authentication management
C	For each automated collection source, the system SHALL maintain: • Collection schedule or subscription configuration • Connection credentials and authentication parameters (stored securely) • Last successful collection timestamp • Collection attempt history and status • Error logs for failed collection attempts
D	The system SHALL implement resilient collection mechanisms: • Retry logic with configurable retry count and backoff intervals • Detection of duplicate data from multiple collection attempts • Graceful handling of temporarily unavailable sources (e.g., retry with configurable backoff, log the failure, and resume automatically when the source becomes available) • Alert generation on persistent collection failures • Automatic recovery and resumption after system restart

NOTE Automated collection sources should be configurable without system restart. The system should support pausing and resuming collection from specific sources for maintenance purposes.

6.3.2.2. Collection Monitoring

Requirement 20	
Identifier	/req/data-ingestion/collection-monitoring

Statement	The system SHALL provide monitoring capabilities for automated data collection: • Dashboard showing status of all configured collection sources • Statistics on collection success/failure rates per source • Data volume metrics (records/observations collected per time period) • Latency metrics (time between observation and ingestion) • Collection log export for auditing and troubleshooting
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6.3.2.3. Data Completeness Monitoring

Requirement 21	
Identifier	/req/data-ingestion/completeness
Statement	The system SHALL support monitoring of data completeness to ensure that all intended observations are received: • Definition and management of expected data: which stations and elements are expected to report, at what frequency, and within what time window • Detection and reporting of missing or late observations against defined expectations • Timely alerting when expected observations fail to arrive within the defined window, to enable corrective action before data becomes unrecoverable • Completeness statistics by station, element, and time period • Distinction between data that was not received and data that was received but failed validation

NOTE

Data completeness monitoring is a distinct concern from collection monitoring (/req/data-ingestion/collection-monitoring), which tracks the technical status of ingestion pipelines (connection success, retry counts, latency). Data completeness monitoring tracks the observation record itself — whether all expected measurements are present and accounted for. NMHSs collecting from numerous stations and communication paths must be able to identify and respond to gaps quickly, before the window for corrective action closes.

6.3.2.4. Manual Data Acquisition

Requirement 22	
Identifier	/req/data-ingestion/manual-acquisition

Statement	The system SHALL support manual data submission including: • Web-based file upload interface, with drag-and-drop support recommended • Batch upload of multiple files • Upload of compressed archives (ZIP, TAR.GZ) • Manual submission via API with authentication • Upload progress indication for large files • Validation feedback during or immediately after upload • Error notification with informative detail in the event of upload failure • Confirmation of successful ingestion with record counts
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6.3.2.5. Format Support

Requirement 23	
Identifier	/req/data-ingestion/formats
Statement	The system SHALL support the ingestion of observational data in the following formats: • CSV and delimited text files: With configurable field mapping, delimiter specification (comma, tab, semicolon, pipe), header row handling, and support for common date/time formats • BUFR: WMO standard binary format (WMO-No. 306, Volume I.2), supporting ingestion of hourly observations, daily summaries, and monthly summaries (including CLIMAT and DAYCLI report types) • WMO text-based formats and code forms: Including FM-12 SYNOP and FM-71 CLIMAT, and other code forms as specified in WMO Manual on Codes (WMO-No. 306, Volume I.1)

NOTE

Supported BUFR report types and WMO code forms should be documented. The Manual on Codes (WMO-No. 306) defines the complete set of applicable report types and code forms.

Recommendation 2	
Identifier	/rec/data-ingestion/formats
Statement	The system SHOULD support the following additional formats: • NetCDF following CF (Climate and Forecast) conventions for station (point) data using discrete sampling geometries • JSON following a defined schema for observational data exchange • XML following a defined schema or profile used in national or regional networks

NOTE

For JSON, XML, and NetCDF formats, a specific schema or profile should be adopted or referenced rather than providing generic format support, as the absence of a defined structure makes generic readers impractical. Support for additional historical or legacy formats may be required based on data rescue activities and national archive requirements; these should be documented in conformance documentation.

6.3.2.6. Format Validation

Requirement 24	
Identifier	/req/data-ingestion/validation
Statement	
A	The system SHALL validate ingested data for: • Format compliance • Required field presence (station ID, timestamp, observed element, value) • Data type correctness (numeric values, valid date/time formats) • Referential integrity (station IDs exist in metadata repository) • Value range plausibility (preliminary sanity checks before full QC)
B	Validation failures SHALL result in: • Rejection at file level as a minimum; record-level rejection SHALL be applied where the format supports it • Generation of detailed error reports identifying specific validation failures • Logging of validation errors for system monitoring and diagnostics

Recommendation 3	
Identifier	/rec/data-ingestion/validation
A	Element-level rejection SHOULD be applied where the format supports it, in preference to record-level rejection.
B	Values that fail the plausibility check SHOULD be stored with a Wrong or invalid quality flag rather than silently discarded. Retaining such values provides evidence of systemic errors in observing or communication systems and supports later investigation.

NOTE

The value range plausibility check in Statement A of /req/data-ingestion/validation is a lightweight gate applied at ingestion time to catch obviously erroneous values (e.g., temperature of 9999°C) before data enters storage. It is not a substitute for the full quality control checks defined in /req/quality-control, which are applied after ingestion using complete station metadata, climatological context, and inter-element relationships. Data that passes ingestion validation but fails full QC will be stored with appropriate quality flags rather than rejected at the ingestion boundary.

6.3.2.7. Data Standardization

Requirement 25	
Identifier	/req/data-ingestion/standardization
Statement	

A	The system SHALL normalize ingested data from all supported formats into a common internal representation that includes: • Observation value with standardized unit of measure • Observation timestamp normalized to UTC • Station identifier (preferentially WIGOS Station Identifier) • Observed element code using WMO BUFR parameter codes or an equivalent controlled vocabulary • Source format identifier • Ingestion timestamp and processing metadata
B	The system SHALL perform unit conversions where necessary, maintaining: • Original value and unit in provenance metadata • Conversion algorithm and factors used • Target standard units as specified in WIGOS metadata standards

NOTE

WMO BUFR parameter codes do not cover all observable elements. The controlled vocabulary used for element codes should be documented, with mappings provided for any elements not covered by the BUFR code tables.

6.3.2.8. Duplicate Detection

Requirement 26	
Identifier	/req/data-ingestion/duplicate-detection
Statement	
A	The system SHALL detect and handle duplicate observations based on: • Station identifier • Sensor identifier (where available) • Observation timestamp • Observed element
B	When duplicates are detected, the system SHALL: • Flag duplicate records for review • Apply configurable duplicate resolution strategies (keep first, keep last, manual review) • Document which duplicate resolution strategy is configured and applied • Log duplicate detection events • Preserve information about all duplicate instances in provenance metadata

NOTE

Operational data exchange systems (including GTS and WIS 2.0) frequently retransmit messages containing identical values. Where duplicate records have identical observed values, automatic suppression without manual review is acceptable. Duplicate resolution requiring manual review should be reserved for cases where values differ between instances.

6.3.2.9. Character Encoding

Requirement 27	
Identifier	/req/data-ingestion/character-encoding
A	The system SHALL handle international character sets: • UTF-8 as default character encoding for all text data • Support for legacy encodings (ASCII, ISO-8859-1) during data ingestion • Proper handling of diacritics and non-Latin scripts in station names and metadata • Character encoding declaration in all text-based exports

6.4. Requirements Class: Data API

Requirements class	/req/data-api
Target type	Software system
Dependency	/req/core, /req/security

6.4.1. Overview

Provide standardized programmatic access to climate observation data and metadata, enabling interoperability with external systems and supporting diverse user communities through well-documented APIs. Standard APIs facilitate integration with WIS 2.0, Regional Climate Centres, and international archives.

6.4.2. Requirements

6.4.2.1. RESTful API

Requirement 28	
Identifier	/req/data-api/rest
Statement	The system SHALL provide a RESTful API that: • Uses HTTP/HTTPS protocols • Implements standard HTTP methods (GET, POST, PUT, DELETE) where appropriate • Returns data in standard formats (JSON, CSV, NetCDF) • Provides comprehensive API documentation • Supports content negotiation based on Accept headers

NOTE

The HTTP PATCH method is not listed as a requirement. Observational records in a CDMS are treated as immutable once ingested; corrections are applied by creating new versioned records with audit trails rather than partially updating existing ones. Implementations that require partial resource updates for non-observational resources (e.g., metadata records) may support PATCH at their discretion.

NOTE

The intended access scope of the API (e.g., internal/LAN access only, or publicly accessible) should be documented. Publicly accessible APIs should apply appropriate authentication and rate limiting in accordance with /req/data-api/auth.

Recommendation 4

Identifier	/rec/data-api/openapi
Statement	API documentation SHOULD be provided in OpenAPI format, enabling interactive documentation and client code generation. OpenAPI specifications should be made available at a well-known endpoint (e.g. /api or /openapi.json).

6.4.2.2. Standards Compliance**Recommendation 5**

Identifier	/rec/data-api/standards
A	The system SHOULD implement one or more of the following standard APIs: • OGC API - Environmental Data Retrieval (EDR) • OGC API - Features • OGC SensorThings API
B	The system SHOULD support XML as a response format, following a defined schema or profile, where required for integration with legacy systems or national and regional networks.

NOTE

Implementation of standardized APIs enhances interoperability with the broader climate and meteorological data ecosystem. The choice of which standards to implement should be based on user community needs and integration requirements.

6.4.2.3. Query Capabilities**Requirement 29**

Identifier	/req/data-api/query
Statement	The system SHALL support API queries with the following parameters: • Spatial extent (bounding box, point, polygon) • Temporal extent (time range, specific timestamp) • Station identifiers (single or multiple) • Observed elements/variables • Quality control status • Aggregation level (raw observations, or pre-computed official statistics where available)

NOTE

On-demand statistical aggregation by the API (e.g., computing daily means from raw observations at query time) introduces risks around data gaps, quality flags, and defined computation periods. Pre-computed official statistics should be stored as standard derived observations (/req/climate-computation/statistics) and served through the same query

interface.

6.4.2.4. Authentication and Authorization

Requirement 30	
Identifier	/req/data-api/auth
Statement	Authentication and authorization SHALL comply with /req/security/authentication and /req/security/authorization

6.4.2.5. API Versioning

Requirement 31	
Identifier	/req/data-api/versioning
Statement	The system SHALL implement API versioning to: • Maintain backward compatibility with existing clients • Clearly indicate API version in endpoints or headers • Provide migration guidance for deprecated API versions • Support multiple API versions concurrently during transition periods

6.4.2.6. Integration Interfaces

Requirement 32	
Identifier	/req/data-api/integration
Statement	The system SHALL provide integration mechanisms for: • WIS 2.0 message broker subscription and publication • Webhook notifications for data availability or quality control events • File-based integration using watched directories or object storage • Message queue integration for asynchronous processing

6.4.2.7. Data Format Interoperability

Requirement 33	
Identifier	/req/data-api/data-formats
Statement	

A	The system SHALL export data in interoperable formats: • NetCDF following CF (Climate and Forecast) conventions for station (point) data using discrete sampling geometries • CSV with RFC 4180 compliance and documented column definitions • JSON following consistent schema conventions • GeoJSON for observations with spatial context • BUFR for exchange with WMO systems, including CLIMAT and DAYCLI message types for monthly and daily summary data
B	Each export format SHALL be accompanied by documentation including: • Description of the data structure and schema • Units of measure for all variables • Quality flag definitions and values • Temporal reference system (UTC or clearly specified) • Spatial reference system (coordinate system, datum)

NOTE

The documentation required in Statement B covers implementation-specific choices (e.g., which NetCDF profile or JSON schema is used, how quality flags are encoded) that are not fully determined by the format standard itself. Machine-readable schemas and format definitions should be made available at a well-known endpoint referenced in the API documentation. For high-frequency or real-time data feeds, it is sufficient to make the format documentation available at this endpoint rather than including it inline with every export response.

6.5. Requirements Class: Basic Climate Computation

Requirements class	/req/climate-computation
Target type	Software system
Dependency	/req/core, /req/quality-control

6.5.1. Overview

Support the computation of derived meteorological variables, statistical aggregates, and homogenized series from quality-controlled observational data. These computed values form the foundation for climate products and long-term climate analysis. This requirements class corresponds to the "Derived observation data" category in WMO-No. 1131 (Section 4.5.1).

6.5.2. Requirements

6.5.2.1. Derived Meteorological Variables

Requirement 34	
Identifier	/req/climate-computation/derived-variables
Statement	

A	The system SHALL compute basic derived meteorological variables from quality-controlled observations where source data do not provide them directly, including at minimum: • Mean sea level pressure derived from station-level pressure, temperature, and station elevation • Dewpoint temperature derived from dry-bulb temperature and wet-bulb temperature or relative humidity • Relative humidity derived from dry-bulb and wet-bulb temperatures or dewpoint temperature
B	Derived values SHALL be stored with provenance metadata recording the source variables used and the computation method applied.

NOTE

Basic derived variables are classified as Required in WMO-No. 1131 (Section 4.5.1.2). Computation formulae are specified in the Guide to Meteorological Instruments and Methods of Observation (WMO-No. 8).

6.5.2.2. Summary Statistics

Requirement 35	
Identifier	/req/climate-computation/statistics
A	The system SHALL support computation and storage of standard summary statistics from quality-controlled observations, including: • Daily summary values: minimum, maximum, mean, and accumulated totals for each observed element • Monthly and annual aggregations of daily summary values • Configurable aggregation periods and functions • Return period estimation for extreme value series (e.g., 10-year, 50-year, 100-year return levels)
B	All computed statistics SHALL be stored as derived observations that: • Reference the source observations used in their computation • Record the computation method, the quality flag thresholds applied, and the proportion of valid source data used • Are clearly distinguished from raw observations in all data access interfaces

NOTE

Rigorous statistical aggregation requires careful handling of data gaps, quality flags, and defined computation periods (which may not align with calendar day boundaries). Statistics computed from series with significant data gaps should be flagged accordingly. Documented minimum data completeness thresholds should be applied before producing summary values.

6.5.2.3. Homogenization: Break Point Detection

Recommendation 6	
Identifier	/rec/climate-computation/homogenization-detection

Statement	The system SHOULD implement statistical methods for detecting inhomogeneities in long-term observation series, supporting: • Relative homogenization using reference series from neighbouring stations • Absolute homogenization methods where appropriate • Detection of multiple break points in a single series • Configurable significance levels for break point detection
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6.5.2.4. Homogenization: Adjustment Calculation

Recommendation 7	
Identifier	/rec/climate-computation/homogenization-adjustment
Statement	The system SHOULD calculate adjustments for detected inhomogeneities: • Compute adjustment factors based on the magnitude of detected breaks • Apply adjustments that preserve long-term means and variability characteristics • Support different adjustment methods (additive, multiplicative, or hybrid) • Validate adjusted series for statistical consistency

NOTE

These homogenization recommendations apply to all observable elements where sufficient station density and record length exist to support homogenization. While temperature is the most commonly homogenized element, the recommendations apply equally to precipitation, pressure, wind, and other elements where applicable.

Homogenization methods are most reliably applied to monthly data. Application to daily or sub-daily data is possible, and important for the assessment of some extremes, but introduces greater uncertainty; the temporal resolution to which homogenization has been applied should be documented.

6.6. Requirements Class: Data Rescue

Requirements class	/req/data-rescue
Target type	Software system
Dependency	/req/core, /req/data-ingestion

6.6.1. Overview

Facilitate the recovery of historical observations from non-digital sources, ensuring rescued data is integrated into the observational records with complete provenance and appropriate quality control. Data rescue extends the temporal coverage of observational records, essential for understanding long-term variability and change.

6.6.2. Requirements

6.6.2.1. Document Imaging Management

Recommendation 8

Identifier	/rec/data-rescue/imaging
Statement	The system SHOULD support document imaging workflows including: • Storage of scanned images with persistent identifiers • Association of images with source document metadata (archive, collection, box number, date range) • Support for multiple image formats (JPEG, TIFF, PDF) • Image quality metadata (resolution, color depth, scan date) • Linkage between images and digitized observations

NOTE

Automated Optical Character Recognition (OCR) of historical documents is outside the scope of this specification. The data entry workflows described here are based on manual keying by operators, with source images displayed alongside data entry forms to support transcription. OCR tools may be used as a supplementary aid at the implementer's discretion.

6.6.2.2. Data Entry Interface

Recommendation 9	
Identifier	/rec/data-rescue/data-entry
A	The system SHOULD provide a data entry interface for manual keying that: • Displays source images alongside data entry forms • Supports configurable data entry templates based on historical form layouts • Provides real-time validation of entered values • Alerts the operator to gross errors that may result from keying mistakes, with the option to re-key the affected values • Tracks data entry operator and timestamp • Supports both single-key and double-key entry workflows for quality assurance • Supports drag-and-drop interaction for assigning images to entry forms, particularly where the variety of historical forms is large

6.6.2.3. Data Rescue Quality Control

Requirement 36	
Identifier	/req/data-rescue/quality-control
Statement	The system SHALL apply quality control to rescued data including: • Where double-key entry is used, comparison between entries to identify transcription errors • Range and consistency checks appropriate to historical observation practices • Verification against original images when discrepancies are detected • Flagging of uncertain or illegible values • Documentation of data rescue quality assurance decisions

NOTE

Digitized observations are processed through the same quality control pipeline as contemporary observations (/req/quality-control). The sequencing of digitization, provisional storage, and quality assurance is an implementation decision; rescued data does not enter the operational observational record without passing through the full QC pipeline.

6.6.2.4. Data Rescue Provenance

Requirement 37	
Identifier	/req/data-rescue/provenance
Statement	The system SHALL record data rescue provenance including: • Source document identification and archive location • Image identifier and scan metadata • Data entry operator(s) and keying date(s) • Keying method used (e.g., single-key, double-key); automated OCR is outside the scope of this specification • Quality assurance verification status • Known limitations or uncertainties in rescued data

6.7. Requirements Class: Metadata Management

Requirements class	/req/metadata-management
Target type	Software system
Dependency	/req/core, /req/security

6.7.1. Overview

Maintain comprehensive metadata supporting data interpretation, quality control, homogenization, and fitness-for-purpose assessment. This encompasses station and sensor metadata (WIGOS), formal definitions of observable elements, dataset-level discovery and lineage metadata, dataset quality reporting, access constraints, and acquisition provenance. Comprehensive metadata is essential for establishing the context and trustworthiness of observational records.

NOTE

Many of the concepts used in this requirements class — including dataset lineage, data quality, access constraints, and feature cataloguing — are in common use in geographic information management. Readers familiar with the ISO 19115 family of standards will recognize these concepts. ISO standards are not freely available and are not requirements of this specification.

6.7.2. Requirements

6.7.2.1. WIGOS Metadata Compliance

Requirement 38	
Identifier	/req/metadata-management/wigos

Statement	
A	The system SHALL support metadata fields as specified in WIGOS Metadata Standard (WMO-No. 1192), including: • Station identification (WIGOS Station Identifier, national identifiers, historical identifiers) • Station name, description, and alternative names • Geographic location (latitude, longitude, elevation, datum reference) • Administrative information (operating organization, country, region) • Station classification (WMO station type, network affiliations, programme participation) • Sensor and instrument inventory with make, model, serial number, and calibration status • Measurement configuration (observed elements, measurement height/depth, sampling interval) • Siting environment description and exposure classification • Operating schedule and reporting practices • Responsible party and contact information
B	The system SHALL support extensibility mechanisms to store metadata fields not covered by the WIGOS Metadata Standard, including national identifiers, legacy identifiers from predecessor networks, and other organization-specific metadata attributes.

6.7.2.2. Station History Documentation

Requirement 39	
Identifier	/req/metadata-management/history
Statement	The system SHALL:
A	maintain a complete station history, including: • Station relocations with old and new coordinates • Instrument changes with make, model, and installation dates • Changes to observation practices or measurement heights • Changes to surrounding environment affecting exposure • Periods of station closure or data gaps • Network affiliation changes
B	for each entry in the station history include: • Effective date and time of the change • Description of the change • Reason for the change where known • Responsible party documenting the change • Supporting documentation references

NOTE

The station history (Statement A) combined with the station and sensor metadata maintained under /req/metadata-management/wigos provides full dataset-to-instrument traceability: for any observation timestamp, the active instrument configuration can be determined from the instrument change records and their effective dates. This information supports reproducibility and uncertainty assessment without requiring a separate acquisition metadata record.

6.7.2.3. Element and Feature Catalogue

Requirement 40	
Identifier	/req/metadata-management/feature-catalogue
Statement	The system SHALL maintain a catalogue of observable elements, recording for each element: • A unique element identifier and human-readable name • Measurement class (e.g., temperature, precipitation, pressure, wind speed) • Accumulation or averaging duration • Unit of measure • Statistical function (instantaneous, mean, maximum, minimum, accumulated total, etc.) • Applicable value range and resolution • Associated controlled vocabulary reference: a WMO BUFR parameter code (from the freely available WMO code tables) or a CF standard name (from the freely available CF Conventions) where applicable

6.7.2.4. Discovery Metadata

Requirement 41	
Identifier	/req/metadata-management/discovery
A	The system SHALL maintain dataset-level discovery metadata records compliant with the WMO Core Metadata Profile version 2 (WCMP2). Each record SHALL include all mandatory WCMP2 properties: • A globally unique persistent identifier for the dataset • Resource or dataset title and free-text description • Spatial extent (bounding box in WGS84) and temporal extent • Earth system discipline classification using the WIS2 Topic Hierarchy vocabulary • Responsible organization name and contact role • WMO data policy classification (core or recommended) • At least one link to a data access endpoint

B	The system SHALL maintain dataset lineage metadata following the W3C PROV Data Model, recording the complete processing history of each dataset. Lineage records SHALL cover: • Source data: originating station networks, instruments, and observation periods • Each significant processing stage applied to the dataset, including at minimum: - Ingestion and format normalization - Quality control procedures and the version or configuration applied - Homogenization and other adjustments, where applicable - Disaggregation or infilling, where applicable - Statistical aggregation or derived product computation, where applicable • For each processing stage: date, description, responsible party, and software or method reference
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NOTE

WCMP2 discovery metadata records are dataset-level descriptions making data findable in WIS2 — they describe what data exists and where to access it. They are distinct from station and sensor metadata (/req/metadata-management/wigos), which describe the observing platform and instruments. For datasets classified as **recommended** under the WMO data policy, at least one access link with associated licence information is required by WCMP2.

Recommendation 10	
Identifier	/rec/metadata-management/gdc-submission
Statement	The system SHOULD support publication of WCMP2-compliant discovery metadata records to the WMO Global Discovery Catalogue via a configured WIS2 Node or equivalent discovery metadata pipeline.

6.7.2.5. Dataset Quality Reporting

Requirement 42	
Identifier	/req/metadata-management/data-quality
A	The system SHALL maintain dataset-level quality metadata including: • Completeness measures: proportion of expected observations present over the dataset period, covering both omission (records absent) and commission (duplicate or spurious records) • Temporal quality: consistency of timestamps and identification of temporal gaps or overlaps • Thematic accuracy: conformance of values to expected ranges and inter-element consistency, drawing on quality control results from /req/quality-control • A conformance result indicating whether the dataset meets defined quality thresholds

B	The system SHALL generate dataset quality reports that: • Summarize completeness by station, element, and time period • Identify temporal gaps and periods of reduced data availability • Report the distribution of quality control flag statuses across the dataset
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NOTE

Dataset-level quality reporting is distinct from per-observation quality flags managed by /req/quality-control. Quality flags on individual observations provide the source data for computing dataset-level quality metrics. These aggregated metrics support fitness-for-purpose assessment, inform users about the suitability of a dataset for specific applications, and should help inform observation providers about any systemic errors or patterns of errors within the dataset.

6.7.2.6. Dataset Maintenance Information

Recommendation 11	
Identifier	/rec/metadata-management/maintenance
Statement	The system SHOULD record dataset maintenance metadata including: • Update frequency (e.g., real-time, daily, monthly, irregular) • Date of the most recent update • Maintenance scope (observations, metadata, quality flags, or derived products) • Responsible party for dataset maintenance • Notes on planned updates or known data gaps

6.7.2.7. Data Access Constraints

Requirement 43	
Identifier	/req/metadata-management/constraints
Statement	The system SHALL record data access and use constraints for each dataset, including: • Legal access constraints: any restrictions on who may access the data • Use constraints: conditions on redistribution or derivative use • Security classification where applicable (e.g., national security requirements) • Embargo periods: dates before which data may not be published externally

Recommendation 12	
Identifier	/rec/metadata-management/constraints
Statement	The system SHOULD record the WMO data policy classification (core or recommended) for each dataset, consistent with the <code>wmo:dataPolicy</code> property specified in WCMP2 and WMO data policy requirements defined in WMO-No. 1238.

NOTE

The constraints in this requirement apply at dataset or network level, not to individual observations.

Constraint metadata must be enforced through the access control mechanisms defined in /req/security/authorization. Embargo periods and redistribution restrictions recorded here should be applied consistently across all data access interfaces, including the API (/req/data-api) and any export functions.

6.7.2.8. Metadata Standards Export

Requirement 44	
Identifier	/req/metadata-management/export
Statement	The system SHALL support export of metadata in standard formats including: • WIGOS Metadata Representation (WMDR) for station and sensor metadata • WMO Core Metadata Profile v2 (WCMP2, GeoJSON) for dataset discovery metadata • W3C PROV-O or equivalent structured format for dataset lineage and provenance records • Dataset quality reports in a documented, structured format (e.g., JSON or XML with defined schema) covering completeness, temporal quality, and quality flag distribution • Element catalogue in a documented structured format referencing WMO BUFR parameter codes or CF standard names • Station metadata exchange formats used by regional or global programmes

NOTE

A simplified second version of the WIGOS Metadata Representation (WMDR) is under development. Further alignment with the updated version is anticipated when published.

6.7.2.9. Metadata Quality Validation

Requirement 45	
Identifier	/req/metadata-management/validation
Statement	The system SHALL
A	validate metadata for: • Required field completeness per WIGOS standards • Coordinate system validity and range checks • Temporal consistency (effective dates do not overlap) • Referential integrity (stations referenced by observations exist and are correct) • Controlled vocabulary compliance for classification fields
B	generate validation reports identifying: • Missing required metadata • Metadata quality issues or inconsistencies • Recommendations for metadata improvement

6.7.2.10. Citation and Attribution

Requirement 46	
Identifier	/req/metadata-management/citation
Statement	The system SHALL support proper data citation by: • Generating unique persistent identifiers (DOIs or equivalents) for datasets • Providing recommended citation text for datasets • Supporting dataset versioning with clear version identifiers • Maintaining links between derived products and source datasets

Recommendation 13	
Identifier	/rec/metadata-management/citation
Statement	The system SHOULD record data usage and access patterns to support impact assessment and reporting on dataset uptake.

6.8. Requirements Class: User Interface

Requirements class	/req/user-interface
Target type	Software system
Dependency	/req/core

6.8.1. Overview

Provide interactive capabilities for scientists, data managers, and users to visualize, explore, and analyze observational data and quality control results. The user interface enables efficient human interaction with complex observational datasets and supports quality assurance workflows.

6.8.2. Requirements

6.8.2.1. Visualization Capabilities

Requirement 47	
Identifier	/req/user-interface/visualization
Statement	The system SHALL provide data visualization capabilities including: • Time series plots for observation values over configurable time ranges • Simultaneous display of original and quality-controlled or homogenized values • Station location maps with filtering and selection capabilities • Ability to export visualizations in common image formats (PNG, SVG, PDF) • Interactive exploration of quality control flags and metadata

6.8.2.2. Tabular Data Display

Requirement 48	
Identifier	/req/user-interface/tabular
Statement	The system SHALL provide tabular data display capabilities including: • Display of observations in tabular format with configurable columns (timestamp, station, element, value, unit, quality flag) • Sorting and filtering by any displayed column • Pagination for large result sets • Simultaneous display of data from multiple stations or elements for comparison • Export of displayed tabular data in CSV or equivalent flat-file format

6.8.2.3. Integrated Search

Requirement 49	
Identifier	/req/user-interface/search
Statement	The system SHALL provide an integrated search interface that: • Supports filtering by station identifier, station name, geographic area, element, time period, and quality control status • Presents search results with sufficient context (station name, location, data coverage) to identify relevant records • Enables navigation from search results to detailed observation views and station metadata • Supports download of search results

6.8.2.4. Data Entry and Forms

Requirement 50	
Identifier	/req/user-interface/data-entry
Statement	The system SHALL provide form-based interfaces supporting: • Manual entry or correction of observation values with real-time validation • Creation and editing of station and sensor metadata records • Review and resolution of quality control flags and alerts • Assignment of quality control decisions with operator identification recorded in the audit trail

NOTE

Data entry forms in the context of this requirements class support operational workflows such as metadata management, manual observation correction, and QC decision recording. Data entry workflows for historical data rescue (keying from scanned documents) are specified separately in /req/data-rescue.

6.9. Requirements Class: Climate Products

Requirements class	/req/climate-products
Target type	Software system
Dependency	/req/core, /req/quality-control, /req/climate-computation

6.9.1. Overview

Support the computation and management of standard climate products from quality-controlled observation series, including climate normals, standard WMO observation products, and climate indices. This requirements class draws on the summary statistics computed by /req/climate-computation. Derived products must be clearly distinguished from raw observations with full documentation of their derivation and data lineage.

6.9.2. Requirements

6.9.2.1. Climate Normals

Requirement 51	
Identifier	/req/climate-products/normals
Statement	The system SHALL support computation and storage of climate normals in accordance with WMO guidelines (WMO-No. 1203), including: • Standard 30-year averaging periods as defined by WMO • Monthly and annual normals for each observed element • Recording of the reference period for each set of normals • Ability to compute and store normals for multiple reference periods simultaneously

6.9.2.2. WMO Standard Observation Products

Requirement 52	
Identifier	/req/climate-products/wmo-products
Statement	The system SHALL support generation of the following WMO standard observation products from quality-controlled observation series: • World Weather Records (WWR): compilation of long-period station records in accordance with WMO-No. 1186, comprising monthly means, extremes, and totals for each observed element over the submission period • CLIMAT messages (FM 71): monthly climatological summaries for submission via WIS 2.0 or other designated exchange mechanisms • DAYCLI messages: daily temperature and precipitation values for submission via WIS 2.0, including required quality flag status and siting classification metadata

NOTE

This requirement covers the generation of these products from quality-controlled observation series held in the CDMS. Ingestion of incoming CLIMAT and DAYCLI

messages from external sources is addressed in /req/data-ingestion/formats.

For DAYCLI, generated messages must include siting classification, measurement quality classification, temperature computation method, and quality control flags for each daily value.

6.9.2.3. Climate Indices

Recommendation 14	
Identifier	/rec/climate-products/indices
Statement	The system SHOULD support computation of standard climate indices, including: • Temperature and precipitation extremes indices for climate change monitoring • Standardized Precipitation Index (SPI) and related drought indices • User-configurable index definitions to accommodate national or regional requirements

6.9.2.4. Product Management

Requirement 53	
Identifier	/req/climate-products/management
Statement	The system SHALL manage derived products such that: • Products are versioned, with earlier versions retained and accessible • Recomputation following correction of source observations generates a new version without overwriting prior versions • Product metadata records the source data period, computation date, and quality constraints applied • Products are accessible through the data API defined in /req/data-api

6.10. Requirements Class: Processing Automation and Scheduling

Requirements class	/req/automation
Target type	Software system
Dependency	/req/core

6.10.1. Overview

Support the scheduled and automated execution of routine data processing workflows, enabling timely and consistent processing without continuous manual intervention. Automation covers quality control processing, statistical product computation, data publication, and system maintenance tasks.

6.10.2. Requirements

6.10.2.1. Scheduled Processing

Requirement 54	
Identifier	/req/automation/scheduling
Statement	The system SHALL support scheduling of routine processing tasks, including: • Time-based scheduling (hourly, daily, monthly, annual, or user-defined intervals) • Event-triggered execution (e.g., automatic quality control following data ingestion) • Configurable processing scope per schedule (station, element, and time period) • Ability to enable and disable schedules without modifying their configuration

6.10.2.2. Processing Workflows

Requirement 55	
Identifier	/req/automation/workflows
Statement	The system SHALL support definition of multi-step processing workflows that: • Chain dependent processing steps in a defined sequence (e.g., quality control → summary statistics → publication) • Halt downstream steps and notify operators when an upstream step fails • Allow manual re-execution of failed or incomplete workflow steps • Support parallel execution of independent processing tasks where applicable

6.10.2.3. Processing Monitoring and Alerting

Requirement 56	
Identifier	/req/automation/monitoring
Statement	The system SHALL provide monitoring and alerting for automated processing, including: • Status visibility of running, completed, and failed tasks • Configurable alerts (email or equivalent notification) when tasks fail or exceed expected duration • Summary view of processing status across all scheduled tasks • Ability for operators to cancel or manually re-trigger processing tasks

6.10.2.4. Processing Audit Log

Requirement 57	
Identifier	/req/automation/audit

Statement	The system SHALL maintain an audit log of all automated processing executions, recording: • Task identifier, type, and configuration at the time of execution • Start time, end time, and completion status • Input data scope: stations, elements, and time period processed • Count of records processed, updated, flagged, or rejected • Any errors or warnings generated during processing
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Chapter 7. Non-functional Requirements

This section defines the quality attributes and constraints that specify how well the Climate Data Management System performs its functions. These requirements address system characteristics essential for operational deployment, long-term sustainability, and user satisfaction.

7.1. Requirements Class: Performance and Scalability

Requirements class	/req/performance
Target type	Software system
Dependency	/req/core

7.1.1. Overview

Ensure the system can handle operational data volumes, concurrent users, and query loads efficiently while maintaining acceptable response times and supporting future growth. Performance requirements ensure the CDMS remains viable as data volumes increase and user communities expand over decades of operation.

7.1.2. Requirements

7.1.2.1. Query Response Performance

Requirement 58	
Identifier	/req/performance/response-time
Statement	The system SHALL: • Define and document response time targets for common query operations, including single-station data retrieval, multi-station spatial queries, and metadata lookups • Monitor query response times against defined targets • Remain responsive under expected concurrent user load without degradation of correctness or data integrity

NOTE	Response time targets should be defined by implementers based on expected user community size, data volumes, and infrastructure. Targets should be documented in system specifications and reviewed as data volumes and user communities grow. Users should be informed of long-running queries through progress indication rather than silent waits.
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7.1.2.2. Data Volume Capacity

Requirement 59	
Identifier	/req/performance/capacity

Statement	The system SHALL: • Support storage and retrieval of observational records spanning multiple decades of historical data across all managed station networks • Accommodate growth in data volume over the operational lifetime of the system without requiring re-architecture of core data storage • Complete ingestion of observations from all configured sources within defined latency targets appropriate to the data delivery schedule of each source
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7.1.2.3. Scalability

Recommendation 15	
Identifier	/rec/performance/scalability
Statement	The system SHOULD be designed to support scaling of processing and storage capacity to accommodate increasing data volumes and user loads as national observing networks grow.

7.2. Requirements Class: Reliability and Availability

Requirements class	/req/reliability
Target type	Software system
Dependency	/req/core

7.2.1. Overview

Ensure the system operates dependably with minimal unplanned downtime, maintains data integrity under all conditions, and recovers gracefully from failures. Reliability is essential for climate data, which serves as an irreplaceable scientific resource requiring protection for future generations.

7.2.2. Requirements

7.2.2.1. System Availability

Requirement 60	
Identifier	/req/reliability/availability
Statement	The system SHALL: • Define and publish availability targets for data retrieval and ingestion services • Monitor system availability and report against defined targets • Communicate scheduled maintenance windows to users in advance • Minimize scheduled maintenance downtime through rolling update or equivalent procedures • Notify administrators promptly of unplanned outages with sufficient detail for diagnosis

NOTE

Retrieval availability is typically held to a higher target than ingestion availability, since brief

ingestion interruptions are often recoverable (source data can be buffered and re-submitted) while retrieval unavailability directly affects users. Implementers should set targets appropriate to their national operational context and infrastructure. Note that availability targets govern day-to-day operations; disaster recovery objectives (RTO/RPO) are addressed separately in /req/reliability/backup-recovery.

7.2.2.2. Data Integrity

Requirement 61	
Identifier	/req/reliability/data-integrity
Statement	The system SHALL ensure data integrity through: • Atomicity of write operations: observation records are written completely or not at all, with no partial writes persisted to storage • Durability of committed records: data written successfully persists across system failures and restarts • Referential integrity constraints preventing orphaned observations (observations without valid station references) • Checksums or cryptographic hash verification for file-based data transfers • Write-ahead logging or equivalent mechanisms to prevent data loss during system failures • Periodic integrity checks to detect and report data corruption

NOTE

Write-ahead logging (WAL) is a standard database durability mechanism in which changes are written to a transaction log before being applied to the main data store. In the event of a system crash, the log enables recovery to a consistent state. The specific mechanism is implementation-dependent; the requirement is that the system can recover to a consistent state following an unplanned failure.

Periodic integrity checks refer to automated verification that stored data matches expected checksums or hash values, detecting silent corruption caused by storage media failures or software defects. Results should be logged and administrators alerted on any detected inconsistency.

7.2.2.3. Backup and Recovery

Requirement 62	
Identifier	/req/reliability/backup-recovery

Statement	The system SHALL implement backup and recovery capabilities: • Automated regular backups of all observation data, metadata, and system configuration • Defined and documented Recovery Time Objective (RTO) and Recovery Point Objective (RPO) appropriate to the operational context • Point-in-time recovery capability to restore the system to a known-good prior state • Documented and regularly tested recovery procedures • Verification of backup integrity through periodic restore testing • Backup data stored in a non-proprietary, documented format to ensure recoverability independent of specific software versions or vendor systems
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NOTE

RTO defines the maximum acceptable time to restore operations following a failure. RPO defines the maximum acceptable data loss measured in time. Targets should be defined appropriate to the operational context — for a national CDMS, a pragmatic RPO will often be longer than for a real-time operational system, since historical observation data is typically re-ingestible from source systems. The critical concern is protecting processed outputs (QC flags, homogenization adjustments) that may not be reproducible without significant effort.

Backup storage volumes grow with retention period. Retention requirements should be balanced against available storage, using incremental backup strategies and compression to manage costs. WMO-No. 1238 requires permanent retention of original and irreproducible climate records and their metadata (3.4.1.4); backup policies should reflect this for long-term archival.

7.2.2.4. Fault Tolerance

Requirement 63	
Identifier	/req/reliability/fault-tolerance
Statement	The system SHALL handle failures gracefully by: • Continuing to serve read requests when write operations are unavailable • Queuing ingestion requests during temporary processing unavailability • Providing meaningful error messages rather than generic failures • Logging all error conditions with sufficient detail for diagnosis • Automatically retrying failed operations with exponential backoff

7.2.2.5. Data Validation on Retrieval

Requirement 64	
Identifier	/req/reliability/retrieval-validation

Statement	The system SHALL validate data consistency when retrieving stored data: • Verify checksums for file-based retrievals • Detect and report any data corruption to system administrators • Provide alternative access to backup copies when primary data is corrupted • Maintain audit logs of data integrity violations
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NOTE

Corruption detection relies on comparing stored checksums or hash values against values recomputed at retrieval time. Alternative access to backup copies implies that backup data is accessible independently of the primary data store; this may be achieved through a secondary system, off-site backup, or disaster recovery capability (see /rec/reliability/disaster-recovery).

7.2.2.6. Disaster Recovery

Recommendation 16	
Identifier	/rec/reliability/disaster-recovery
Statement	The system SHOULD support disaster recovery capability including: • A secondary system or equivalent standby capable of assuming operations following catastrophic failure of the primary system • Documented switchover procedures, including manual activation steps and validation checks to confirm data integrity following switchover • Replication or backup transfer to a geographically separate location • Regular testing of disaster recovery procedures to verify recoverability

NOTE

A disaster recovery capability (sometimes termed a cold standby or Disaster Recovery Centre) provides resilience against catastrophic failure of the primary infrastructure (fire, flood, hardware failure). It is distinct from high-availability clustering, which addresses routine failures within the primary site. Switchover procedures should be documented in system maintenance documentation as required by /req/maintainability/documentation.

7.3. Requirements Class: Security

Requirements class	/req/security
Target type	Software system
Dependency	/req/core

7.3.1. Overview

Protect climate data assets, ensure appropriate access controls, maintain confidentiality where required, and provide comprehensive audit trails of security-relevant events. Security safeguards the integrity and trustworthiness of the climate record while enabling appropriate access for legitimate users.

7.3.2. Requirements

7.3.2.1. Authentication

Requirement 65	
Identifier	/req/security/authentication
Statement	The system SHALL implement authentication mechanisms that: • Support integration with institutional or enterprise identity management systems using standard authentication protocols • Enforce strong credential policies for locally managed accounts • Support multi-factor authentication (MFA) for administrative and privileged accounts • Implement account lockout after repeated failed authentication attempts • Maintain secure session management with automatic timeout after inactivity

NOTE

Standard authentication protocols include OpenID Connect, OAuth 2.0, SAML 2.0, and LDAP, among others. The choice of protocol should reflect the identity management infrastructure in use within the deploying organization. OpenID Connect is widely adopted and provides a standards-based federated identity layer suitable for both web and API access.

7.3.2.2. Authorization and Access Control

Requirement 66	
Identifier	/req/security/authorization
Statement	The system SHALL implement role-based access control (RBAC) that: • Separates privileges across user roles including at minimum: read-only users, data entry operators, quality control analysts, and system administrators • Is configurable to restrict data access at station or network level • Enforces data access policies, especially for embargoed or sensitive datasets • Applies the principle of least privilege: users are granted the minimum permissions necessary for their role

Recommendation 17	
Identifier	/rec/security/authorization
Statement	The system SHOULD allow system administrators to define and manage custom roles and their associated permissions, in addition to the pre-defined roles required above.

NOTE

Role definitions and their associated permissions should be configurable by system administrators without requiring code changes or system restart. Access restrictions at station or network level are important for national systems where data from certain networks may be subject to bilateral sharing agreements or national security constraints.

7.3.2.3. Data Encryption

Requirement 67	
Identifier	/req/security/encryption
Statement	The system SHALL implement encryption for: • All data in transit using current industry-standard transport security protocols • Sensitive data at rest including user credentials, API keys, and personal information • Backup files stored on external or off-site media

7.3.2.4. Audit Logging

Requirement 68	
Identifier	/req/security/audit-logging
Statement	The system SHALL:
A	maintain comprehensive audit logs recording: • All user authentication attempts (successful and failed) • All data and metadata modifications (create, update, delete) with user identification • All administrative operations (configuration changes, user management, permission changes) • All quality control flag overrides with justification • All data export operations including user, timestamp, and data selection criteria
B	Audit logs SHALL: • Be tamper-proof or stored in write-once media • Be retained in accordance with the organization's data retention policy • Be searchable and exportable for security incident investigation • Include sufficient detail to reconstruct the sequence of events

7.3.2.5. Vulnerability Management

Requirement 69	
Identifier	/req/security/vulnerability-management
A	The system SHALL support security maintenance through: • Regular security updates to address known vulnerabilities • Documented process for applying security patches • Vulnerability scanning capability or support for external security assessment tools • Security advisories for known issues affecting the deployment

7.3.2.6. Data Privacy

Requirement 70	
Identifier	/req/security/data-privacy
A	The system SHALL protect personal information by: • Minimizing collection and storage of personally identifiable information (PII) to what is operationally necessary • Providing mechanisms to redact or anonymize personal information where required • Supporting compliance with applicable data protection regulations as determined by the deploying organization • Documenting data retention policies for any personal data held by the system

NOTE

The applicable data protection regulations will vary by jurisdiction. Implementers should assess which regulations apply to their deployment (e.g., national privacy laws, regional frameworks) and configure the system accordingly. The primary personal data held by a CDMS is typically contact information for station operators, data submitters, data monitors, quality control operators, and data extraction users, rather than observation data itself.

7.3.2.7. Provenance Security

Requirement 71	
Identifier	/req/security/provenance-security
A	The system SHALL protect provenance records through: • Cryptographic timestamps or equivalent mechanisms to prevent backdating • Tamper-evident storage for provenance audit trails

7.4. Requirements Class: Usability

Requirements class	/req/usability
Target type	Software system
Dependency	/req/core

7.4.1. Overview

Ensure the system is accessible, intuitive, and efficient for climate scientists, data managers, and operators with varying levels of technical expertise. Usability directly impacts the effectiveness of quality control, data validation, and metadata management workflows that depend on expert human judgment.

7.4.2. Requirements

7.4.2.1. User Interface Design

Requirement 72

Identifier	/req/usability/ui-design
A	The system user interface SHALL: • Provide consistent navigation patterns across all functions • Use clear, unambiguous labels and terminology aligned with climate science conventions • Provide contextual help and documentation accessible from relevant screens • Support keyboard navigation for efficiency • Indicate required fields and validation requirements clearly before submission • Provide meaningful error messages with guidance on corrective actions

NOTE

A web-based user interface is the recommended approach as it maximizes accessibility across platforms, reduces client-side deployment complexity, and facilitates integration with web-based identity management systems. Web-based interfaces should apply the accessibility recommendations in /rec/usability/accessibility.

7.4.2.2. Accessibility

Recommendation 18	
Identifier	/rec/usability/accessibility
A	The system user interface SHOULD conform to: • Web Content Accessibility Guidelines (WCAG) 2.1 Level AA where applicable • Support for screen readers and assistive technologies • Sufficient color contrast for visual accessibility • Text scaling without loss of functionality • Keyboard-only operation for all critical functions

7.4.2.3. User Feedback and Notifications

Requirement 73	
Identifier	/req/usability/notifications
A	The system SHALL provide notifications for: • Completion of long-running operations (ingestion, QC processing, statistical computation) • Critical quality control failures requiring manual review • System maintenance schedules • Data embargo or access restriction changes • Configurable notification delivery methods (email, in-application alerts)

7.5. Requirements Class: Maintainability

Requirements class	/req/maintainability
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Target type	Software system
Dependency	/req/core

7.5.1. Overview

Ensure the system can be updated, configured, and troubleshot efficiently with minimal disruption to operations. Maintainability is critical for long-term sustainability, as climate data systems must operate reliably over multiple decades with evolving technologies and requirements.

7.5.2. Requirements

7.5.2.1. Configuration Management

Requirement 74	
Identifier	/req/maintainability/configuration
A	The system SHALL support configuration management with: • Externalized configuration files separate from application code • Hot-reload capability for quality control threshold updates without system restart • Configuration version control with rollback capability • Configuration validation before activation to prevent invalid settings • Documentation of all configurable parameters with valid value ranges

7.5.2.2. Logging and Diagnostics

Requirement 75	
Identifier	/req/maintainability/logging
A	The system SHALL provide logging and diagnostic capabilities: • Configurable log levels (ERROR, WARN, INFO, DEBUG) per component • Structured logging with timestamps, component identification, and correlation IDs • Log rotation and retention policies to manage disk space • Centralized logging capability for distributed deployments • Performance metrics logging (response times, throughput, resource utilization)

7.5.2.3. Health Monitoring

Requirement 76	
Identifier	/req/maintainability/health-monitoring

A	The system SHALL expose health monitoring endpoints providing: • Overall system health status (healthy, degraded, unhealthy) • Component-level health status for all major system components • Key performance indicators (query response times, ingestion rates, error rates) • Resource utilization metrics (CPU, memory, disk, network) • Integration with standard monitoring protocols or tools used within the deployment environment
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7.5.2.4. API Backward Compatibility

Requirement 77	
Identifier	/req/maintainability/api-compatibility
A	The system SHALL maintain API stability through: • Semantic versioning of API endpoints • A defined and published deprecation policy specifying the support period for deprecated API versions • Clear deprecation notices in API documentation and response headers • Migration guides for breaking changes • Ability to run multiple API versions concurrently during transition periods

7.5.2.5. Database Schema Evolution

Requirement 78	
Identifier	/req/maintainability/schema-evolution
A	The system SHALL support database schema changes with: • Version-controlled schema migration scripts • Automated or documented migration procedures • Ability to test schema migrations in non-production environments • Rollback capability for failed migrations • Backward compatibility for schema changes where feasible

7.5.2.6. Documentation

Requirement 79	
Identifier	/req/maintainability/documentation

A	The system SHALL provide comprehensive documentation including: • System architecture and component descriptions • Installation and deployment procedures • Configuration reference with parameter descriptions and examples • API reference documentation with request/response examples • Troubleshooting guides for common issues • Backup and recovery procedures • Disaster recovery and system switchover procedures • Security hardening guidelines
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7.6. Requirements Class: Interoperability

Requirements class	/req/interoperability
Target type	Software system
Dependency	/req/core, /req/data-api

7.6.1. Overview

Ensure the system can exchange data and integrate with external systems using industry-standard protocols, formats, and conventions. Interoperability enables climate data to flow seamlessly between national systems, regional centers, international archives, and the WIS 2.0 infrastructure.

7.6.2. Requirements

7.6.2.1. Standards Compliance

Requirement 80	
Identifier	/req/interoperability/standards
A	The system SHALL comply with relevant international standards: • WMO WIGOS Metadata Standard for station metadata • WMO Manual on Codes (WMO-No. 306) for BUFR and code form support • CF Conventions for NetCDF climate data files • OGC standards (API - Features, API - EDR, API - Records, or SensorThings API) for data access and discovery APIs

7.6.2.2. API Interoperability

Requirement 81	
Identifier	/req/interoperability/api

A	The system API SHALL support interoperability through: • Content negotiation using standard HTTP Accept headers • Standard HTTP status codes with appropriate semantics • CORS (Cross-Origin Resource Sharing) support for web-based clients • OpenAPI (Swagger) specification or equivalent machine-readable API description
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7.7. Requirements Class: Data Governance

Requirements class	/req/data-governance
Target type	Software system
Dependency	/req/core

7.7.1. Overview

Ensure proper stewardship of climate data through lifecycle management, quality documentation, access policies, and compliance with organizational and regulatory requirements. Data governance provides the policy framework and accountability structures necessary for long-term climate data stewardship.

7.7.2. Requirements

7.7.2.1. Data Lifecycle Management

Requirement 82	
Identifier	/req/data-governance/lifecycle
Statement	The system SHALL support data lifecycle management with: • Configurable data retention policies per dataset type or network • Automated archival of historical data to long-term storage • Data deletion workflows with approval requirements for permanent deletion • Tombstone records for deleted data maintaining metadata about what was deleted and why • Migration paths for data format updates or schema changes

7.7.2.2. Data Quality Metadata

Requirement 83	
Identifier	/req/data-governance/quality-metadata

Statement	The system SHALL maintain data quality metadata including: • Completeness metrics (percentage of expected observations received) • Timeliness metrics (delay between observation time and system receipt) • Quality control statistics (pass/fail rates per check type) • Manual intervention frequency and types • Homogenization status and adjustment statistics • Overall fitness-for-purpose assessments
B	Quality metadata SHALL be: • Queryable alongside observation data • Exportable in machine-readable formats • Updated automatically as data quality changes

NOTE

Fitness-for-purpose refers to an assessment of whether a dataset meets the quality requirements of a particular application. For example, a dataset with frequent short gaps may be suitable for climatological trend analysis but unsuitable for computing climate normals. Fitness-for-purpose assessments summarize quality metrics (completeness, temporal continuity, QC flag distribution) in terms of suitability for defined use cases, supporting informed data selection by users. See also [/req/metadata-management/data-quality](#).

7.7.2.3. Data Access Policies

Requirement 84	
Identifier	/req/data-governance/access-policies
Statement	The system SHALL enforce data access policies including: • Public/restricted/confidential data classification • Embargo periods for recently collected data where required • License and attribution requirements for data usage • Data sharing agreements and restrictions per network or organization • Automated enforcement of access restrictions based on user roles

7.7.2.4. Data Stewardship Roles

Requirement 85	
Identifier	/req/data-governance/stewardship

Statement	The system SHALL define and support data stewardship roles including: • Data custodian responsibilities for dataset maintenance • Data quality manager authority for quality control processes, including initiating investigations to resolve identified data quality issues (e.g. systematic errors or dataflow problems) • Data access coordinator for managing access requests and policies • System administrator separation from data stewardship roles • Clear escalation paths for data governance issues
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NOTE

Data stewardship roles define the organizational responsibilities for data management and are distinct from the technical access control roles defined in /req/security/authorization. In practice, stewardship roles map to RBAC roles: for example, the data quality manager role is associated with permissions to override quality control flags and approve homogenization adjustments. The RBAC configuration in the security class provides the technical enforcement of the responsibilities defined here.

Appendix A: Requirements Summary

This annex provides a consolidated reference list of all normative requirements and recommendations defined in this specification, grouped by requirements class.

A.1. Functional Requirements

Identifier	Type	Requirements Class
/req/core/data-model	Requirement	Core Data Management
/req/core/observation-storage	Requirement	Core Data Management
/req/core/observation-retrieval	Requirement	Core Data Management
/req/core/metadata-storage	Requirement	Core Data Management
/req/core/metadata-retrieval	Requirement	Core Data Management
/req/core/metadata-versioning	Requirement	Core Data Management
/req/core/provenance-storage	Requirement	Core Data Management
/req/core/provenance-retrieval	Requirement	Core Data Management
/req/core/provenance-immutability	Requirement	Core Data Management
/req/core/batch-processing	Requirement	Core Data Management
/req/core/timezone	Requirement	Core Data Management
/req/quality-control/observation-retention	Requirement	Observations Quality Control
/req/quality-control/flags	Requirement	Observations Quality Control
/req/quality-control/configuration	Requirement	Observations Quality Control
/req/quality-control/execution	Requirement	Observations Quality Control
/req/quality-control/range-checks	Requirement	Observations Quality Control
/req/quality-control/temporal-checks	Requirement	Observations Quality Control

Identifier	Type	Requirements Class
/req/quality-control/spatial-checks	Requirement	Observations Quality Control
/req/quality-control/internal-consistency	Requirement	Observations Quality Control
/rec/quality-control/flagging-convention	Recommendation	Observations Quality Control
/req/data-ingestion/automated-collection	Requirement	Data Ingestion
/req/data-ingestion/collection-monitoring	Requirement	Data Ingestion
/req/data-ingestion/completeness	Requirement	Data Ingestion
/req/data-ingestion/manual-acquisition	Requirement	Data Ingestion
/req/data-ingestion/formats	Requirement	Data Ingestion
/req/data-ingestion/validation	Requirement	Data Ingestion
/req/data-ingestion/standardization	Requirement	Data Ingestion
/req/data-ingestion/duplicate-detection	Requirement	Data Ingestion
/req/data-ingestion/character-encoding	Requirement	Data Ingestion
/rec/data-ingestion/formats	Recommendation	Data Ingestion
/rec/data-ingestion/validation	Recommendation	Data Ingestion
/req/data-api/rest	Requirement	Data API
/req/data-api/query	Requirement	Data API
/req/data-api/auth	Requirement	Data API
/req/data-api/versioning	Requirement	Data API
/req/data-api/integration	Requirement	Data API
/req/data-api/data-formats	Requirement	Data API

Identifier	Type	Requirements Class
/rec/data-api/openapi	Recommendation	Data API
/rec/data-api/standards	Recommendation	Data API
/req/climate-computation/derived-variables	Requirement	Basic Climate Computation
/req/climate-computation/statistics	Requirement	Basic Climate Computation
/rec/climate-computation/homogenization-detection	Recommendation	Basic Climate Computation
/rec/climate-computation/homogenization-adjustment	Recommendation	Basic Climate Computation
/rec/data-rescue/imaging	Recommendation	Data Rescue
/rec/data-rescue/data-entry	Recommendation	Data Rescue
/req/data-rescue/quality-control	Requirement	Data Rescue
/req/data-rescue/provenance	Requirement	Data Rescue
/req/metadata-management/wigos	Requirement	Metadata Management
/req/metadata-management/history	Requirement	Metadata Management
/req/metadata-management/feature-catalogue	Requirement	Metadata Management
/req/metadata-management/discovery	Requirement	Metadata Management
/req/metadata-management/data-quality	Requirement	Metadata Management
/req/metadata-management/constraints	Requirement	Metadata Management
/req/metadata-management/export	Requirement	Metadata Management
/req/metadata-management/validation	Requirement	Metadata Management
/req/metadata-management/citation	Requirement	Metadata Management
/rec/metadata-management/gdc-submission	Recommendation	Metadata Management

Identifier	Type	Requirements Class
/rec/metadata-management/maintenance	Recommendation	Metadata Management
/rec/metadata-management/constraints	Recommendation	Metadata Management
/rec/metadata-management/citation	Recommendation	Metadata Management
/req/user-interface/visualization	Requirement	User Interface
/req/user-interface/tabular	Requirement	User Interface
/req/user-interface/search	Requirement	User Interface
/req/user-interface/data-entry	Requirement	User Interface
/req/climate-products/normals	Requirement	Climate Products
/req/climate-products/wmo-products	Requirement	Climate Products
/req/climate-products/management	Requirement	Climate Products
/rec/climate-products/indices	Recommendation	Climate Products
/req/automation/scheduling	Requirement	Processing Automation
/req/automation/workflows	Requirement	Processing Automation
/req/automation/monitoring	Requirement	Processing Automation
/req/automation/audit	Requirement	Processing Automation

A.2. Non-functional Requirements

Identifier	Type	Requirements Class
/req/performance/response-time	Requirement	Performance
/req/performance/capacity	Requirement	Performance
/rec/performance/scalability	Recommendation	Performance

Identifier	Type	Requirements Class
/req/reliability/availability	Requirement	Reliability and Availability
/req/reliability/data-integrity	Requirement	Reliability and Availability
/req/reliability/backup-recovery	Requirement	Reliability and Availability
/req/reliability/fault-tolerance	Requirement	Reliability and Availability
/req/reliability/retrieval-validation	Requirement	Reliability and Availability
/rec/reliability/disaster-recovery	Recommendation	Reliability and Availability
/req/security/authentication	Requirement	Security
/req/security/authorization	Requirement	Security
/req/security/encryption	Requirement	Security
/req/security/audit-logging	Requirement	Security
/req/security/vulnerability-management	Requirement	Security
/req/security/data-privacy	Requirement	Security
/req/security/provenance-security	Requirement	Security
/rec/security/authorization	Recommendation	Security
/req/usability/ui-design	Requirement	Usability
/req/usability/notifications	Requirement	Usability
/rec/usability/accessibility	Recommendation	Usability
/req/maintainability/configuration	Requirement	Maintainability
/req/maintainability/logging	Requirement	Maintainability
/req/maintainability/health-monitoring	Requirement	Maintainability

Identifier	Type	Requirements Class
/req/maintainability/api-compatibility	Requirement	Maintainability
/req/maintainability/schema-evolution	Requirement	Maintainability
/req/maintainability/documentation	Requirement	Maintainability
/req/interoperability/standards	Requirement	Interoperability
/req/interoperability/api	Requirement	Interoperability
/req/data-governance/lifecycle	Requirement	Data Governance
/req/data-governance/quality-metadata	Requirement	Data Governance
/req/data-governance/access-policies	Requirement	Data Governance
/req/data-governance/stewardship	Requirement	Data Governance

Appendix B: Acknowledgements

This specification was developed under an informal working group of the Expert Team on Information Management (ET-IM), operating under the Standing Committee on Information Management and Technology (SC-IMT) of the Commission for Observation, Infrastructure and Information Systems (INFCOM) of the World Meteorological Organization (WMO). The work was carried out in close collaboration with the Services Commission (SERCOM), bringing together CDMS developers and climate domain experts to ensure the requirements are both technically sound and operationally grounded.

The following individuals contributed to the development and review of this specification:

Chair: Ian Edwards (myAcorn, UK)

Members and contributors: Etna Cervantes (MCH, Mexico), Marcellin Habimana (Climsoft, Kenya), Charlotte McBride (South African Weather Service), Patrick Munyoki (Climsoft, Kenya), Ge Peng (USA), Xavier Perrot (Météo-France International), Dwayne Scott (Belize Meteorological Service / SURFACE CDMS), Denis Stuber (Météo-France), Peter Thorne (Maynooth University, Ireland), William Wright (Australia), Liu Yiming (China Meteorological Administration)

WMO Secretariat: David Berry, Washington Otieno

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